

# Combined forces of hydrostatic pressure and actin polymerization drive endothelial tip cell migration and sprouting angiogenesis

Reviewed Preprint

v1 • July 29, 2024

Not revised

Igor Kondrychyn, Haymar Wint, Liqun He, Christer Betsholtz, Li-Kun Phng 

Laboratory for Vascular Morphogenesis, RIKEN Center for Biosystems Dynamics Research, Kobe 650-0047, Japan • Department of Immunology, Genetics and Pathology, Rudbeck Laboratory, Uppsala University, 75185 Uppsala, Sweden • Department of Medicine Huddinge, Karolinska Institutet, 14157 Huddinge, Sweden

 [https://en.wikipedia.org/wiki/Open\\_access](https://en.wikipedia.org/wiki/Open_access)
 Copyright information

## Abstract

### Summary

Cell migration is a key process in the shaping and formation of tissues. During sprouting angiogenesis, endothelial tip cells invade avascular tissues by generating actomyosin-dependent forces that drive cell migration and vascular expansion. Surprisingly, ECs can still invade if actin polymerization is inhibited. In this study, we show that endothelial tip cells employ an alternative mechanism of cell migration that is dependent on Aquaporin (Aqp)-mediated water inflow and increase in hydrostatic pressure. In the zebrafish, ECs express *aqp1a.1* and *aqp8a.1* in newly formed vascular sprouts in a VEGFR2-dependent manner. Aqp1a.1 and Aqp8a.1 loss-of-function studies show an impairment in intersegmental vessels formation because of a decreased capacity of tip cells to increase their cytoplasmic volume and generate membrane protrusions, leading to delayed tip cell emergence from the dorsal aorta and slower migration. Further inhibition of actin polymerization resulted in a greater decrease in sprouting angiogenesis, indicating that ECs employ two mechanisms for robust cell migration *in vivo*. Our study highlights an important role of hydrostatic pressure in tissue morphogenesis.

#### eLife assessment

This study **convincingly** shows that aquaporins play a key role in blood vessel formation during zebrafish development. In particular, the paper implicates hydrostatic pressure and water flow as mechanisms controlling endothelial cell migration during angiogenic sprouting. This **important** study significantly advances our understanding of cell migration during morphogenesis. As such, this work will be of great interest to developmental and cell biologists working on organogenesis, angiogenesis, and cell migration.

<https://doi.org/10.7554/eLife.98612.1.sa3>

## Introduction

The blood vascular system is a dynamic, multicellular tissue that adapts to the metabolic demands of development, growth and homeostasis by altering its pattern and morphology. Macroscopic transformation in the vascular network is achieved by microscopic changes in endothelial cell (EC) behaviours. For example, vascular expansion through sprouting angiogenesis requires the coordination of EC migration, proliferation, anastomosis, lumen formation and cell rearrangements, all of which require specific cell shape changes. At the initial phase of sprouting angiogenesis, endothelial tip cells generate numerous membrane protrusions such as filopodia and lamellipodia that drive migration and anastomosis (Figueiredo et al., 2021 [DOI](#); Gerhardt et al., 2003 [DOI](#); Phng et al., 2013 [DOI](#); Sauter et al., 2017 [DOI](#)). During tubulogenesis, apical membranes deform by generating inverse blebs to expand lumens so that blood can be transported efficiently through the blood vascular network (Gebala et al., 2016 [DOI](#)). However, following vessel perfusion, ECs become less dynamic and instead develop resistance to the deforming forces of blood pressure to maintain vessel morphology (Kondrychyn et al., 2020 [DOI](#)). EC shape is therefore dynamic and changes depending on the morphogenetic state of vascular development.

It is well established that localized remodelling of actin cytoskeleton and non-myosin II contractility are instrumental in driving cell shape changes during tissue morphogenesis (Clarke and Martin, 2021 [DOI](#); Munjal and Lecuit, 2014 [DOI](#); Murrell et al., 2015 [DOI](#)). More recently, there is an accumulating body of work implicating a role of water flow and the resulting changes in hydrostatic pressure as another mechanical mechanism of cell shape change (Choudhury et al., 2022 [DOI](#); Chugh et al., 2022 [DOI](#); Li et al., 2020 [DOI](#)). For example, in the osmotic engine model of cell migration, water inflow at the leading edge and outflow at the rear leads to forward translocation of tumour cells in confined microenvironment (Stroka et al., 2014 [DOI](#)). Additionally, hydrostatic pressure from the extracellular environment can drive tissue morphogenesis, as demonstrated during mouse blastocyst formation (Chan et al., 2019 [DOI](#); Dumortier et al., 2019 [DOI](#)), development of the otic vesicle in the zebrafish (Mosaligni et al., 2019 [DOI](#)) and blood vessel lumen expansion (Gebala et al., 2016 [DOI](#)). Hydrostatic pressure can therefore generate sufficient forces to shape tissues and organs to their proper form and size during development.

We have previously discovered that EC migration persists after the inhibition of actin polymerization and in the absence of filopodia to generate new blood vessels in the zebrafish (Phng et al., 2013 [DOI](#)). This observation suggests the existence of an alternative mechanism of migration independent of actin polymerization. In this study, we sought to determine whether hydrostatic pressure regulates EC migration during sprouting angiogenesis in the zebrafish by investigating the function of Aquaporins (Aqp), which are transmembrane water channels that increase water permeability of cell membranes to promote transcellular water flow (Day et al., 2014 [DOI](#); Farinas et al., 1997 [DOI](#); Kozono et al., 2002 [DOI](#); Preston et al., 1992 [DOI](#)). We discovered that ECs of newly formed intersegmental vessels (ISV) express *aqp1a.1* and *aqp8a.1* mRNA, and observed the enrichment of Aqp1a.1 and Aqp8a.1 proteins in the leading edge of migrating tip cells. Detailed single cell analyses showed that endothelial tip cells lacking *aqp1a.1* and *aqp8a.1* expression have reduced cell volume, membrane protrusions and migration capacity. As a result, there is defective sprouting angiogenesis and ISV formation in *aqp1a.1;aqp8a.1* double mutant zebrafish. Notably, when actin polymerization is inhibited in *aqp1a.1;aqp8a.1* double mutant zebrafish, there is a greater impairment in EC migration and sprouting angiogenesis, demonstrating the additive function of actin polymerization and hydrostatic pressure (that is created by water inflow) in generating membrane protrusions to drive EC migration *in vivo*.

## Results

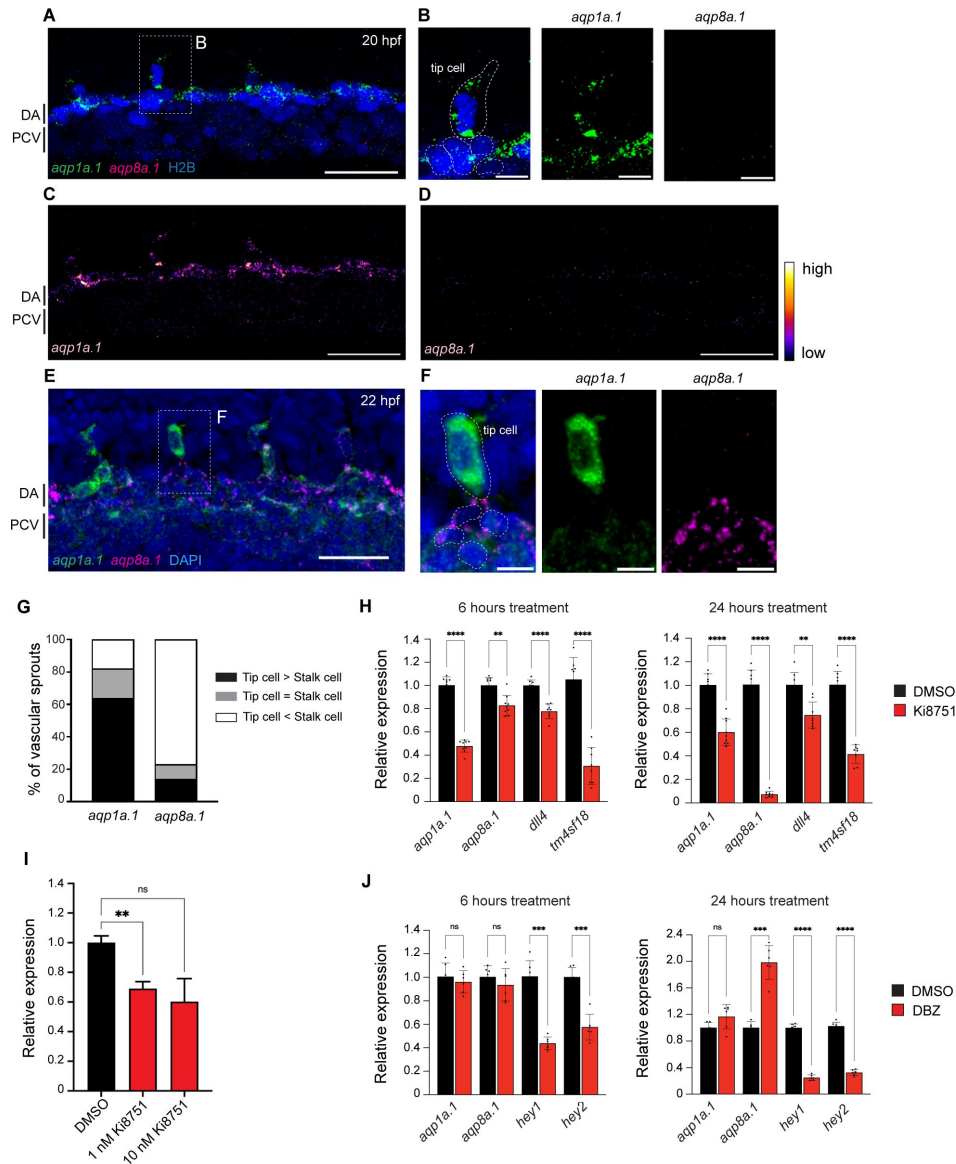
### ***aqp1a.1* and *aqp8a.1* are differentially expressed by tip and stalk cells during sprouting angiogenesis**

Of the 18 number of *aquaporin* genes that are expressed in zebrafish (Tingaud-Sequeira et al., 2010), *aqp1a.1* and *aqp8a.1* mRNA have been detected in blood endothelial cells (ECs) (Koun et al., 2016; Rehn et al., 2011). We confirmed the endothelial expression of *aqp1a.1* and *aqp8a.1* at 30 hours post fertilization (hpf) and 2-, 3- and 4-days post fertilization (dpf) by whole-mount *in situ* hybridization (Fig. S1). The two endothelial *aqp* genes, however, differ in spatial expression. While *aqp1a.1* is widely expressed in blood vessels of the head and trunk (dorsal aorta (DA), caudal artery (CA), ISVs, dorsal longitudinal anastomotic vessels (DLAV) and caudal vein plexus) at 30 hpf, *aqp8a.1* expression is absent in cerebral blood vessels and is restricted to the DA, CA and the ventral regions of ISVs in the trunk. After 1 dpf, *aqp8a.1* expression expands to the entire ISV and DLAV. Additionally, the expression of *aqp1a.1* and *aqp8a.1* gradually decreases in the DA and CA so that both are absent by 4 dpf. Single cell RNA sequencing (scRNAseq) of ECs isolated at 34 hpf and 3 dpf further confirmed the endothelial expression of *aqp1a.1* and *aqp8a.1* mRNA (Fig. S2 A - D), revealed differential expression of *aqp1a.1* and *aqp8a.1* mRNA in distinct endothelial clusters and highlighted higher *aqp1a.1* transcript expression in all endothelial subtypes compared to *aqp8a.1* (Fig. S2 E - H).

We next examined the expression pattern of *aqp1a.1* and *aqp8a.1* at the beginning of sprouting angiogenesis at higher resolution using RNAscope and confocal microscopy. At 20 hpf, *aqp1a.1* mRNA is highly expressed in the DA with little expression in the PCV (Fig. 1A, C and Fig. S3). Closer inspection uncovered heterogeneous expression in the DA, with higher expression in ECs that would be specified as tip cells (Fig. 1A, C and Fig. S3B). Indeed, tip cells of newly formed vascular sprouts express higher levels of *aqp1a.1* compared to adjacent ECs in the DA at 20 hpf (Fig. 1A – C, Fig. S3B) and the majority of ISVs (61%) display higher *aqp1a.1* expression in tip cells than in trailing stalk cells at 22 hpf (Fig. 1G). Although we previously observed *aqp8a.1* mRNA in the DA at 30 hpf (Fig. S1), its expression here is largely absent at 20 hpf (Fig. 1B, D and Fig. S3C). At 22hpf, *aqp8a.1* mRNA expression in the DA increases (Fig. 1E and F). Interestingly, unlike *aqp1a.1*, in 78% of ISVs analyzed, *aqp8a.1* mRNA is more highly expressed in stalk cells than tip cells (Fig. 1G). These observations highlight differential expression pattern of *aqp1a.1* and *aqp8a.1* in newly formed vascular sprouts, with *aqp1a.1* expression enriched in tip cells and *aqp8a.1* expression in stalk cells.

### **VEGF-VEGFR2 signalling induces the expression of *aqp1a.1* and *aqp8a.1* mRNA**

As *aqp1a.1* and *aqp8a.1* mRNA display differential tip-stalk cell expression, we explored whether their expression is regulated by the VEGFR2-Notch signalling axis, which controls endothelial tip-stalk cell specification during sprouting angiogenesis (Benedito and Hellström, 2013; Phng and Gerhardt, 2009; Siekmann et al., 2013). To inhibit VEGFR2 activity, embryos were treated with 1  $\mu$ M ki8751 from 20 hpf. Relative quantitative polymerase chain reaction (qPCR) experiments showed that *aqp1a.1* mRNA expression was suppressed by 52% after 6 hours ( $p < 0.0001$ ) and by 40% after 24 hours ( $p < 0.0001$ ) of ki8751 treatment when compared to DMSO-treated embryos (Fig. 1H). *aqp8a.1* mRNA expression was also decreased by the inhibition of VEGFR2 activity, and its suppression was accentuated after 24 hours treatment (93% decrease,  $p < 0.0001$ ) compared to after 6 hours (17% decrease,  $p = 0.0028$ ). These results demonstrate that VEGFR2 activity upregulates the expression of *aqp1a.1* and *aqp8a.1*, but the temporal dynamics is different for *aqp1a.1* and *aqp8a.1*. To examine whether VEGFR2 activity also regulates Aquaporin



**Figure 1.**

### Differential expression of *aqp1a.1* and *aqp8a.1* mRNAs in tip and stalk cells.

(A-F) Detection of *aqp1a.1* and *aqp8a.1* mRNA by RNA scope *in situ* hybridization in 20 (A - D) and 22 (E and F) hpf zebrafish embryos. Representative maximum intensity projection of confocal z-stacks of tip and stalk cells sprouting from DA are shown (n = 9 embryos, 2 independent experiments). B and F show magnified region of a tip cell from a 20 and 22 hpf zebrafish, respectively (tip cell and EC nuclei in the DA are outlined). Scale bar, 10  $\mu$ m (B and F) and 40  $\mu$ m (A, C - E). Percentage of vascular sprouts with differential expression of *aqp1a.1* and *aqp8a.1* mRNAs in tip and stalk cells in 22 hpf zebrafish (n=23 sprouts from 8 embryos, 2 independent experiments). (G). qPCR analysis of *aqp1a.1* and *aqp8a.1* expression in zebrafish embryos treated with 0.01% DMSO or 1  $\mu$ M ki8751 at 20 hpf for 6 or 24 hours (n = 3 independent experiments). (G) qPCR analysis of *AQP1* gene expression in HAECs treated with 0.01% DMSO or 1 nM and 10 nM ki8751 for 6 hr (n = 2 independent experiments). (J) qPCR analysis of *aqp1a.1* and *aqp8a.1* gene expression in the zebrafish embryos treated with 0.1% DMSO or 10  $\mu$ M DBZ at 20 hpf for 6 or 24 hr (n = 2 independent experiments). In panels H-J, gene expressions are shown relative to *gapdh* expression and data are presented as mean  $\pm$  SD; statistical significance was determined by Brown-Forsythe and Welch ANOVA tests with Dunnett's multiple comparisons test; ns  $p > 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , and \*\*\*\* $p < 0.0001$ . DA, dorsal aorta; PCV, posterior cardinal vein. See also Fig. S3.

expression in human ECs, we treated cultured human aortic endothelial cells (HAECs) with ki8751 for 6 hours and analyzed *AQP1* expression by qPCR. *AQP8* expression was not examined since it is not expressed in HAECs (data not shown). Inhibition of VEGFR2 activity with 1 nM and 10 nM ki8751 decreased *AQP1* mRNA expression by 31% ( $p = 0.0023$ , **Fig. 1I**) and 40% ( $p = 0.1032$ , **Fig. 1I**), respectively, demonstrating cross-species conservation of VEGFR2 activity in inducing *Aqp1* mRNA expression.

Notch signalling was inhibited using the gamma-secretase inhibitor, Dibenazepine (DBZ), which prevents proteolytic cleavage and consequently, activation of the Notch receptor (Groth et al., 2010). Treatment of embryos with DBZ from 20 hpf for 6 and 24 hours significantly decreased the expression of *hey1* and *hey2*, canonical targets of Notch signalling (**Fig. 1J**), confirming the effectiveness of the drug treatment in our experiments. When compared to DMSO-treated embryos, 6-hour treatment with DBZ did not cause a significant change in *aqp1a.1* and *aqp8a.1* expression levels (**Fig. 1J**). Longer treatment with DBZ for 24 hours, however, increased *aqp8a.1* expression by 98% ( $p < 0.0001$ ) while *aqp1a.1* expression was relatively unchanged (**Fig. 1J**). These findings suggest that Notch signalling inhibits *aqp8a.1*, but not *aqp1a.1*, expression.

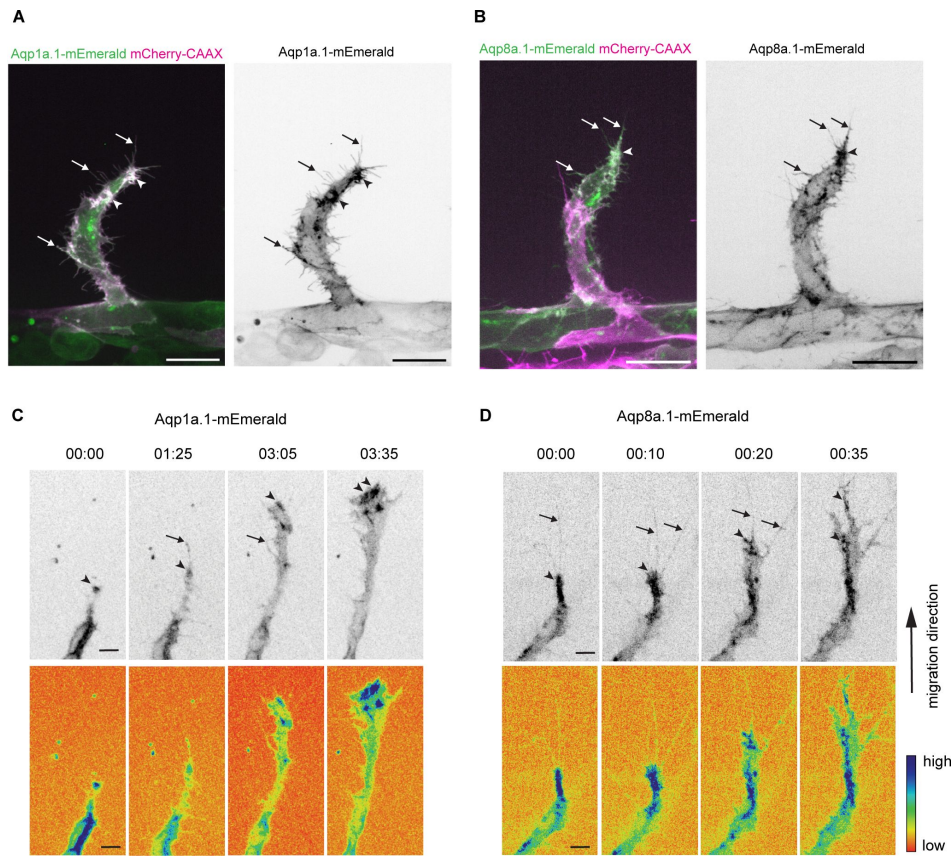
## Aqp1a.1 and Aqp8a.1 proteins are enriched at the leading front of migrating tip cells

To gain a better understanding of where within the cell Aquaporin protein functions, we generated stable *Tg(fli1ep:aqp1a.1-mEmerald)<sup>rk30</sup>* and *Tg(fli1ep:aqp8a.1-mEmerald)<sup>rk31</sup>* zebrafish lines in which ECs express Aqp1a.1 or Aqp8a.1 tagged with mEmerald, respectively. At 25 hpf, we observed an accumulation of Aqp1a.1 (**Fig. 2A**) and Aqp8a.1 (**Fig. 2B**) in mCherryCAAX-positive membrane clusters in the cytoplasm and filopodia of tip cells. Interestingly, many of the Aqp1a.1- or Aqp8a.1-positive membrane clusters are located at the migrating front of the tip cell. Time-lapse imaging capturing the process of ISV formation between 24 to 30 hpf further showed sustained enrichment of Aqp1a.1 and Aqp8a.1 in clusters at the front of the tip cell as well as in filopodia (**Fig. 2C and D**, Movie S1 and S2), suggesting that Aqp1a.1 and Aqp8a.1 may promote localized water flux at the migrating edge of tip cells.

## Loss of Aqp1a.1 and Aqp8a.1 function leads to defective trunk blood vessel formation

We subsequently proceeded to generate zebrafish *aqp1a.1* and *aqp8a.1* mutants to investigate the function of Aquaporin-mediated water flow in blood vessel development. Using the CRISPR/Cas9 genome editing technique, we targeted the first exons of *aqp1a.1* and *aqp8a.1* genes. The CRISPR-generating allele *aqp1a.1<sup>rk28</sup>* carries a 14-bp deletion and an 8-bp insertion in the 5' end of the gene leading to a premature termination codon at amino acid 76 after 1 missense amino acid (**Fig. S4A-B**). The *aqp8a.1<sup>rk29</sup>* allele carries a 4-bp deletion in the 5' end of the gene that leads to a frameshift after T15 and premature termination codon at amino acid 73 after 58 missense amino acids (**Fig. S5A-B**). Premature termination codons are often more deleterious than missense mutations because they result in the loss of protein expression. Nonsense-mediated mRNA decay is a quality-control mechanism that selectively degrades mRNAs harboring premature termination codons (Chang et al., 2007). Using whole-mount *in situ* hybridization and qPCR analysis, we demonstrated that *aqp1a.1* and *aqp8a.1* mRNA expression is significantly decreased in *aqp1a.1<sup>rk28/rk28</sup>* (**Fig. S4C-D**) and *aqp8a.1<sup>rk29/rk29</sup>* (**Fig. S5C-D**) zebrafish embryos, respectively, suggesting a rapid degradation of mutant mRNAs and as a result, the loss of Aquaporin protein expression.

Although *aqp1a.1<sup>rk28/rk28</sup>*; *aqp8a.1<sup>rk29/rk29</sup>* embryos look morphologically normal when compared to wildtype or *aqp1a.1<sup>+/rk28</sup>*; *aqp8a.1<sup>+/rk29</sup>* embryos (**Fig. S8A, B, C and D**), we observed defects in blood vessel formation and patterning at 1 to 3 dpf. Compared to 28 hpf control embryos whose



**Figure 2.**

### Aquaporin proteins are enriched at the leading front of migrating cells.

(A and B) Representative maximum intensity projection confocal z-stacks of endothelial tip cells of 25 hpf *Tg(fli1ep:aqp1a.1-mEmerald)<sup>rk30</sup>;(kdrl:ras-mCherry)<sup>s916</sup>* (A) and *Tg(fli1ep:aqp8a.1-mEmerald)<sup>rk31</sup>;(kdrl:ras-mCherry)<sup>s916</sup>* (B) embryos. Scale bar, 20  $\mu$ m. (C and D) Still images from time-lapse movies of migrating tip cells from *Tg(fli1ep:aqp1a.1-mEmerald)<sup>rk30</sup>* (C) and *Tg(fli1ep:aqp8a.1-mEmerald)<sup>rk31</sup>* (D) embryos. Movies were taken from 24 to 30 hpf. Black arrowheads, Aquaporin protein localization at the leading edge of migrating tip cells. Arrows, Aquaporin localization in filopodia. Time, h:min. Scale bar, 5  $\mu$ m. See also Movie S1 and S2.

ISVs have reached the dorsal roof of the neural tube and started to form the DLAV, ISVs of *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos are much less developed (**Fig. 3A and D**) with significantly reduced length (**Fig. 3E** and Fig. S6H and K). Single deletion of *aqp1a.1* (**Fig. 3B** and Fig. S6F and I) or *aqp8a.1* (**Fig. 3C** and Fig. S6G and J) also led to defective ISV formation, although the defects are less severe compared to the loss of both *aqp1a.1* and *aqp8a.1* expression. Comparison of ISV length in *aqp1a.1<sup>rk28/rk28</sup>*, *aqp8a.1<sup>rk29/rk29</sup>* and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos shows that the loss of *aqp1a.1* leads to a greater decrease in ISV length than the loss of *aqp8a.1* (**Fig. 3E**, Fig. S6I and J), and that the loss of both *aqp1a.1* and *aqp8a.1* resulted in the greatest reduction. These observations indicate an additive effect of Aqp1a.1 and Aqp8a.1 function in blood vessel development and implicate Aqp1a.1 as the more dominant Aquaporin protein in EC function, in line with the higher expression of *aqp1a.1* mRNA in tip cells.

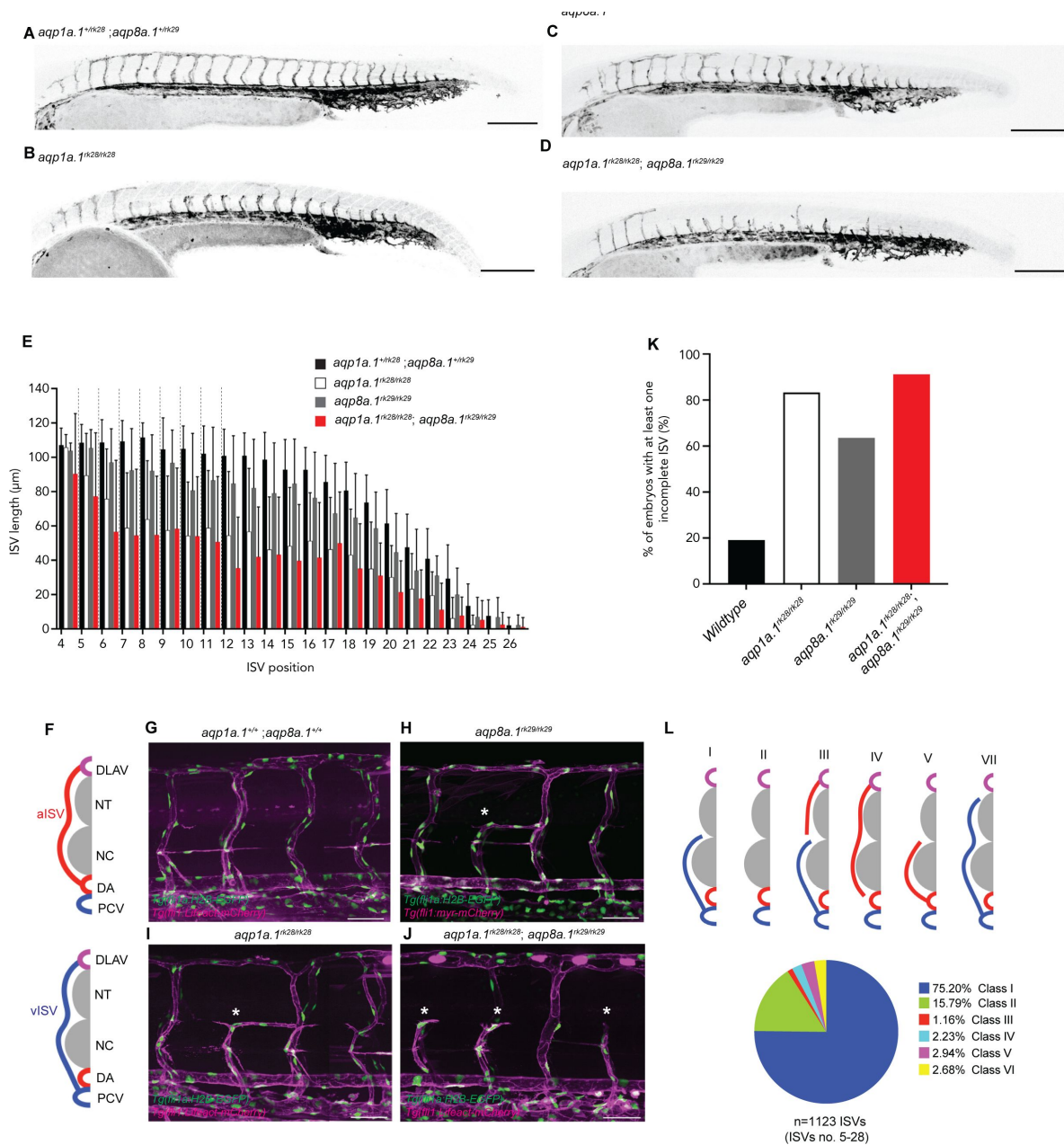
We next examined the embryos at 3 dpf, a stage when arterial ISVs (aISVs) and venous (vISVs) are fully established and lumenized, connecting the DLAV to either the DA or PCV (**Fig. 3F and G**), respectively. Such fully established ISVs are significantly reduced in embryos when *aqp1a.1*, *aqp8a.1* or both *aqp1a.1* and *aqp8a.1* are deleted. Instead, these embryos display an increased number of aISVs and vISVs that fail to establish a connection to the DLAV, DA or PCV (**Fig. 3H – J**). Quantification shows that while 19.1% of wildtype embryos display at least 1 incomplete ISV per embryo, this increased to 65.8% in *aqp8a.1<sup>rk29/rk29</sup>* embryos, 83.3% in *aqp1a.1<sup>rk28/rk28</sup>* embryos and 92.2% in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos (**Fig. 3K**). The number of incomplete ISVs per embryo in *aquaporin* mutants was variable, between 1 to 6, 1 to 16 and 1 to 22 incomplete ISVs in *aqp8a.1<sup>rk29/rk29</sup>*, *aqp1a.1<sup>rk28/rk28</sup>* and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos, respectively (Fig. S6A–D). We also found that incomplete ISVs appear with higher frequency in the zebrafish trunk (ISV number 6 to 15) rather than in the tail (ISV number 16 to 27, Fig. S6E). Detailed analysis revealed that most of the incomplete ISVs are vISVs that are connected to the PCV but not the DLAV (Class I, 75.2%), followed by the absence of aISV or vISV (Class II, 15.8%, **Fig. 3L** and Fig. S7A and B). A small fraction of the incompletely formed vessels are aISVs that are not connected to the DA (Class IV, 2.2%, Fig. S7D) or DLAV (Class V, 2.9%, Fig. S7E), or vISVs that are not connected to the DLAV (Class VII, 2.68%, Fig. S7F). At some regions, there is an aISV that is not connected to the DA and a vISV that is not connected to the DLAV (Class III, 1.2%, Fig. S7C).

As *aqp1a.1* is expressed in brain vasculature during development (Fig. S1), we examined whether cerebral vasculature is also affected in *aqp1a.1<sup>rk28/rk28</sup>* embryos. At 4 dpf, we found that the length of the dorsal longitudinal vein (DLV), a main blood vessel that supplies blood to the choroid plexus (Bill and Korzh, 2014) was reduced in *aqp1a.1<sup>rk28/rk28</sup>* embryos ( $p = 0.0098$ , Fig. S9B). Also, in 2 out of 21 *aqp1a.1<sup>rk28/rk28</sup>* embryos, the DLV was undeveloped (Fig. S9A and C), a defect that is also observed in 50% of *vegfab<sup>-/-</sup>* and *vegfc<sup>-/-</sup>* mutant zebrafish (Parab et al., 2021).

In summary, our analyses suggest that in the absence of Aqp1a.1 and/or Aqp8a.1 function, there is an initial delay in ISV formation during primary sprouting angiogenesis. By 3 dpf, some ISVs are completely formed (Fig. S8K and M), connecting the DA or PCV to the DLAV. However, the number of completely formed ISVs is significantly decreased, with the loss of Aqp1a.1 having a more severe impact compared to the loss of Aqp8a.1, and the combined loss of Aqp1a.1 or Aqp8a.1 having a greater deleterious effect.

## Aquaporins promote endothelial tip cell protrusion and migration

To gain a better understanding of how incomplete ISV formation arise, we performed timelapse imaging in zebrafish depleted of both Aqp1a.1 and Aqp8a.1 function from 21 – 22 hpf, when sprouting angiogenesis begins. In *aqp1a.1<sup>+rk28</sup>;aqp8a.1<sup>+rk29</sup>* embryos, tip cells emerge from the DA between 20 and 24 hpf (**Fig. 4A**) and migrate dorsally between the somite boundaries over the notochord and neural tube to form primary ISVs (**Fig. 4B**, Movie S3). However, in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos, tip cell nuclei emerge between 22 and 28 hpf (**Fig. 4A**). Furthermore, in some *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos, tip cells retract whereby



**Figure 3.**

### Loss of Aquaporin function leads to defective trunk vessel formation.

(A-D) Representative maximum intensity projection confocal z-stacks of 28 hpf *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (A), *aqp1a.1<sup>rk28/rk28</sup>* (B), *aqp8a.1<sup>rk29/rk29</sup>* (C) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (D) embryos. (E) Quantification of ISV length in 28 hpf embryos (*aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>*: n = 21 embryos; *aqp1a.1<sup>rk28/rk28</sup>*: n = 20 embryos; *aqp8a.1<sup>rk29/rk29</sup>*: n = 20 embryos; *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>*: n = 19 embryos from 2 independent experiments). (F) Illustration of aISV and vISV connections in the trunk of wild type embryo at 3 dpf. (G-J) Representative maximum intensity projection confocal z-stacks of 3 dpf wild type (G), *aqp8a.1<sup>rk29/rk29</sup>* (H), *aqp1a.1<sup>rk28/rk28</sup>* (I), and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (J) embryos. Asterisk marks incomplete ISVs. (K) Quantification of percentage of wild type and *aquaporin* mutant embryos at 3 dpf with at least one incomplete ISVs per embryo (wild type, n = 47 embryos; *aqp1a.1<sup>rk28/rk28</sup>*, n = 30 embryos; *aqp8a.1<sup>rk29/rk29</sup>*, n = 38 embryos; *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>*, n = 204 embryos). (L) A pie chart showing proportion of different classes of incomplete ISV phenotypes (I-VI) found in *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* mutant embryos at 3 dpf. DA, dorsal aorta; DLAV, dorsal longitudinal anastomotic vessel; ISV, intersegmental vessel; aISV, arterial ISV; vISV, venous ISV; NC, notochord; NT, neural tube; PCV, posterior cardinal vein. Scale bar, 50 μm (G-J) and 200 μm (A-D). See also Fig. S6, S7 and S8.

membrane protrusions do not extend dorsally in the direction of migration but withdraw into the DA so that a primary ISV does not form at this location (Fig. 4C, Movie S4). These observations indicate that tip cell protrusion from the DA is significantly delayed in the absence of Aqp1a.1 and Aqp8a.1 function.

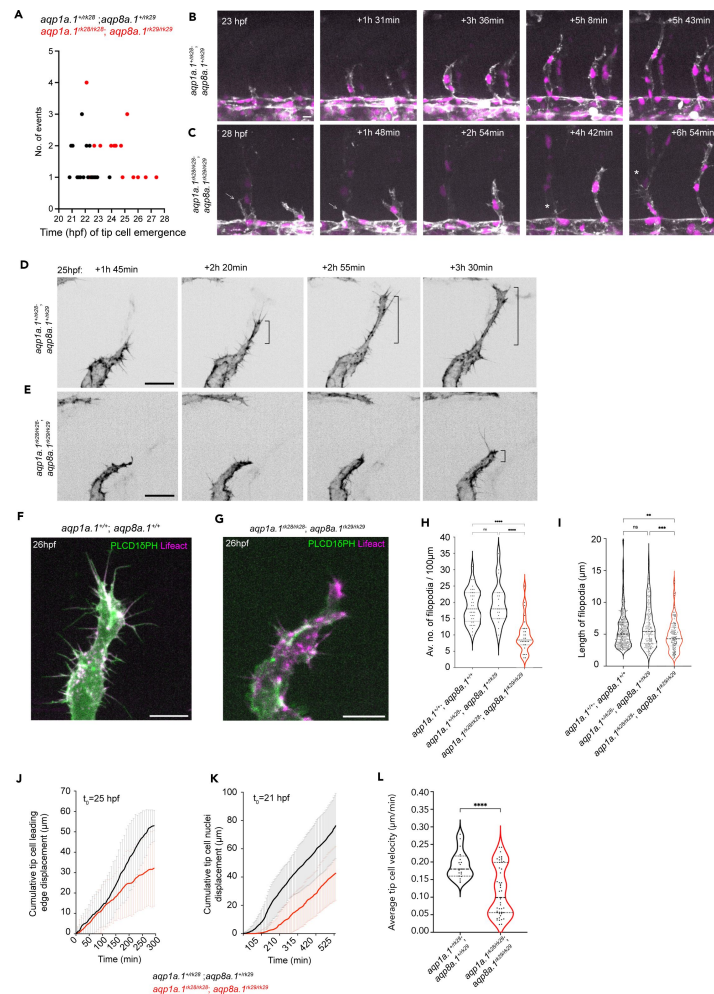
We have previously demonstrated that filopodia facilitate EC migration by serving as a template for lamellipodia formation (Phng et al., 2013). Lamellipodia frequently emanate laterally from stable filopodia at the leading edge of tip cells, leading to rapid expansion of the filopodium, increase in volume and the formation of a stable protrusion in the direction of migration (Phng et al., 2013). This was similarly observed in *aqp1a.1<sup>+/rk28</sup>;aqp8a.1<sup>+/rk29</sup>* embryos (Fig. 4D, Movie S5) but not in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos (Fig. 4E, Movie S6). Comparison of filopodia dynamics in wildtype (Fig. 4F), *aqp1a.1<sup>+/rk28</sup>;aqp8a.1<sup>+/rk29</sup>* and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* (Fig. 4G) embryos revealed a significant reduction in filopodia number (wildtype,  $18.75 \pm 5.1/100\mu\text{m}$ ; *aqp1a.1<sup>+/rk28</sup>;aqp8a.1<sup>+/rk29</sup>*,  $19.8 \pm 6.3/100\mu\text{m}$ ; *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>*,  $9.9 \pm 5.2/100\mu\text{m}$ ; Fig. 4H) and length (wildtype,  $5.9 \pm 3.2 \mu\text{m}$ ; *aqp1a.1<sup>+/rk28</sup>;aqp8a.1<sup>+/rk29</sup>*,  $6.2 \pm 3.3 \mu\text{m}$ ; *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>*,  $4.8 \pm 2.4 \mu\text{m}$ ; Fig. 4I) in ECs lacking both Aqp1a.1 and Aqp8a.1 function. Analysis of time-lapse images of tip cell behaviour further showed slower tip cell membrane expansion at the leading edge (Fig. 4J), decreased nuclei displacement (Fig. 4K) and a significant reduction in tip cell migration velocity (Fig. 4L) in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos.

In conclusion, Aqp1a.1 and Aqp8a.1 are required for sprouting angiogenesis by promoting endothelial tip cell emergence from the DA, the formation of membrane protrusions and endothelial cell migration.

## Tip cell volume regulation depends on Aquaporin-mediated water influx

Aquaporin channels permit bidirectional flow of water across the plasma membrane, with the direction of flow determined by the osmotic gradient between the extracellular environment and cell cytoplasm (Agre et al., 2002). We next sought to determine whether water flows in or out of tip cells during sprouting angiogenesis. We hypothesize that water flux across the cell membrane can lead to changes in tip cell volume, with water influx increasing the volume while outflow decreases. To quantify tip cell volume, we injected a plasmid encoding *kdrl:mEmerald* transgene into 1-cell stage wildtype or *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos to label ECs in a mosaic manner. mEmerald-positive tip cells were imaged and their volume measured at 24 to 25 hpf. Wildtype ECs are on average  $946.6 \pm 299.8 \mu\text{m}^3$  in size (Fig. 5A and C). In the absence of both Aqp1a.1 and Aqp8a.1 function (Fig. 5B), tip cell volume significantly decreased by about 24% to  $718.4 \pm 308.5 \mu\text{m}^3$  ( $p = 0.0176$ , Fig. 5C) when compared to wildtype tip cells. This finding supports water influx as the direction of water flow in tip cells during migration, and that this is mediated through Aqp1a.1 and Aqp8a.1.

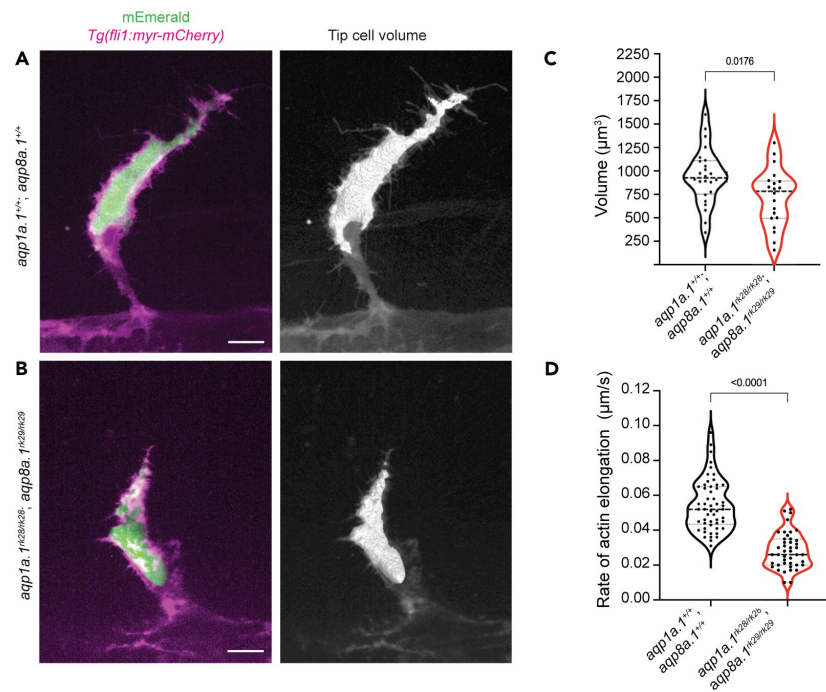
It has been proposed that water influx through Aquaporin increases the diffusion of actin monomers and increases actin polymerisation (Loitto et al., 2007; Papadopoulos et al., 2008). Given that filopodia formation is dependent on actin polymerization (Mallavarapu and Mitchison, 1999; Ridley, 2011), we examined whether actin polymerization is perturbed in the absence of Aqp1a.1 and Aqp8a.1 by quantifying the growth rate of actin bundles in tip cell filopodia by tracking Lifeact-mCherry signal in the filopodia. In wildtype embryos, the average growth rate of actin is  $0.054 \pm 0.015 \mu\text{m}$  per second whereas in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos, the growth rate is significantly decreased by 47% to  $0.028 \pm 0.010 \mu\text{m}$  per second ( $p < 0.0001$ , Fig. 5D). Our observations therefore support a role of zebrafish Aqp1a.1 and Aqp8a.1 in promoting actin polymerisation and consequently, filopodia number and length, which are reduced in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos (Fig. 4F and G).



**Figure 4.**

### Aquaporins promote endothelial tip cell protrusion and migration.

(A) Timing of tip cell emergence from the DA in *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (n = 23) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 27). (B and C) Still images from time-lapse imaging of migrating tip cells from *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (B) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (C) embryos from 20 to 30 hpf. White arrows, retracting tip cell. \*, secondary sprouting from PCV. Scale bar, 10  $\mu$ m. (D and E) Stills images from representative time-lapse movies of migrating tip cells in *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (D, n = 2) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (E, n = 5) embryos. Movies were taken from 25 to 30 hpf (n = 2 independent experiments). Bracket, formation of stable protrusions. Scale bar, 20  $\mu$ m. (F and G) Representative maximum intensity projection confocal z-stacks of tip cells from wild type (F) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (G) embryos at 26 hpf. Scale bar, 10  $\mu$ m. (H) Quantification of filopodia number in tip cells of wild type (n = 36 cells from 10 embryos, 2 independent experiments), *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (n = 19 cells from 6 embryos, 2 independent experiments) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 28 cells from 9 embryos, 2 independent experiments) embryos. (I) Quantification of filopodia length in tip cells of wild type (n = 24 cells from 7 embryos, 2 independent experiments), *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (n = 16 cells from 7 embryos, 2 independent experiments) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 11 cells from 6 embryos, 2 independent experiments) embryos. (J) Quantification of tip cell leading edge displacement of *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (n = 19 cells from 5 embryos, 3 independent experiments) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 47 cells from 13 embryos, 6 independent experiments) embryos at 25 - 30 hpf. (K) Quantification of tip cell nuclei displacement of *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (n = 20 cells from 6 embryos, 4 independent experiments) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 20 cells from 4 embryos, 2 independent experiments) embryos at 21 - 30 hpf. (L) Quantification of tip cell migration velocity in *aqp1a.1<sup>+/rk28</sup>; aqp8a.1<sup>+/rk29</sup>* (n = 19 cells from 5 embryos, 3 independent experiments) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 47 cells from 13 embryos, 6 independent experiments) embryos at 25 - 30 hpf. Statistical significance was determined by Brown-Forsythe and Welch ANOVA tests with Dunnett's (H) or Sidak's (I) multiple comparisons test, and with unpaired t-test (L). ns,  $p > 0.05$ ; \*,  $p < 0.05$ ; \*\*\*,  $p < 0.001$ ; \*\*\*\*,  $p < 0.0001$ . See also Movie S3-6.



**Figure 5.**

**Aquaporin-mediated water influx increases tip cell volume and actin elongation in filopodia.**

(A and B) Representative maximum intensity projection confocal z-stacks of tip cell and its 3D-volume rendering of 25 hpf wild type (A) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (B) labelled with cytosolic mEmerald. Scale bar, 10 μm. (C) Quantification of tip cell volume in 25 hpf wild type (n = 24 cells) and *aqp1a.1<sup>rk28/rk28</sup>; aqp8a.1<sup>rk29/rk29</sup>* (n = 20 cells) embryos from 3 independent experiments. (D) Quantification of growth rate of actin bundles in tip cell filopodia in 25 hpf wild type (n = 12 cells from 7 embryos, 2 independent experiments) and *aqp1a.1<sup>rk28</sup>; aqp8a.1<sup>rk29</sup>* (n = 12 cells from 6 embryos, 2 independent experiments) embryos. Statistical significance was determined by unpaired *t*-test with (C) or without (D) Welch's correction.

In summary, our results demonstrate a role of endothelial Aquaporins in increasing water influx and as a consequence, endothelial tip cell volume and hydrostatic pressure, as well as in promoting actin polymerization.

## Additive effects of actin polymerization and water influx in driving EC migration and sprouting angiogenesis

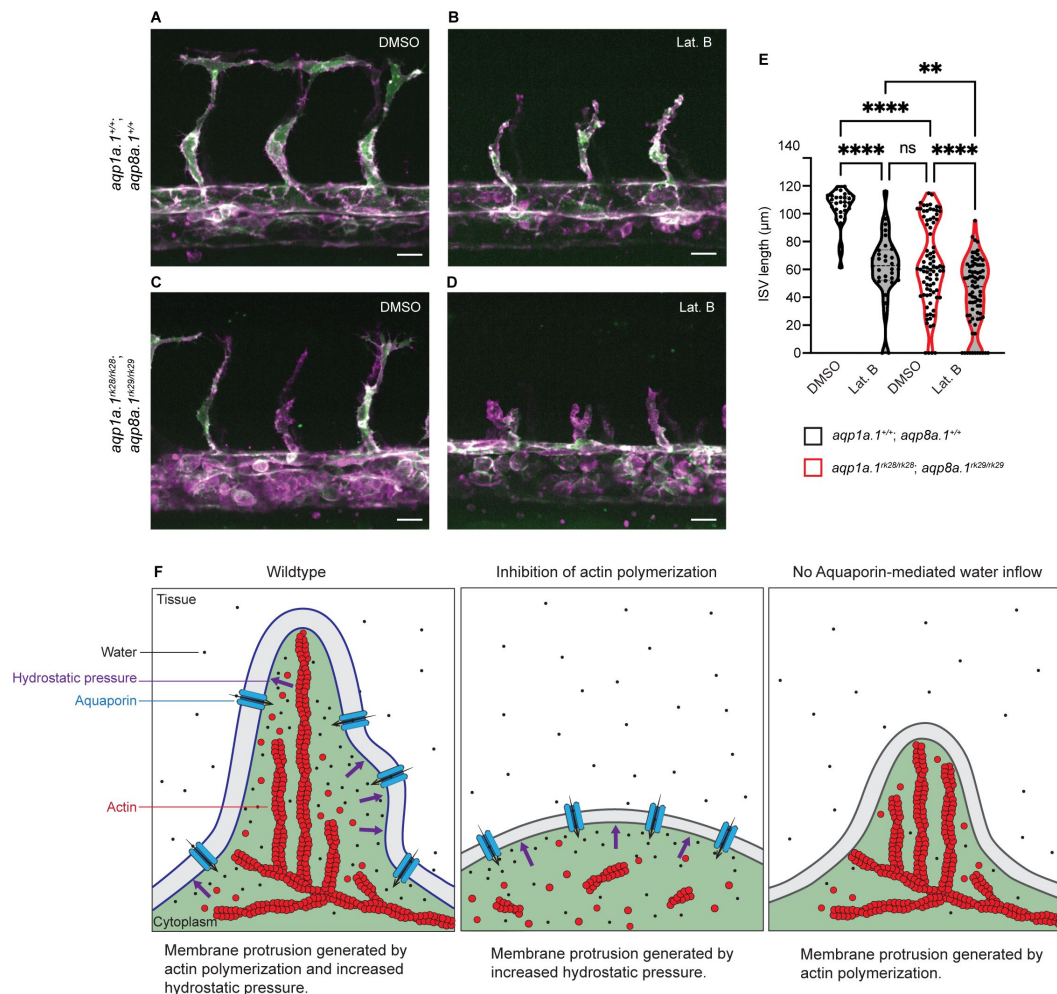
We have previously observed that ECs are still able to migrate to generate ISVs after the inhibition of actin polymerization (Phng et al., 2013). In these experiments, embryos were treated with a low concentration of Latrunculin B (Lat. B), an inhibitor of actin polymerization, that resulted in the suppression of filopodia formation. Under such conditions, ECs can generate small membrane protrusions and migrate more slowly in a directed manner to generate ISVs. How ECs are still able to migrate after the inhibition of actin polymerization was unclear. As our current study demonstrates a role of water inflow and hydrostatic pressure as another mechanism of cell migration, we hypothesize that ECs employ water influx to migrate when actin polymerization is compromised and that the depletion of both water influx and actin polymerization will result in a greater inhibition of EC migration.

To test this hypothesis, we examined ISV development in embryos with decreased actin polymerization and water inflow by treating *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos with Lat. B. At 28 hpf, ISVs of wildtype embryos treated with DMSO have reached the dorsal roof of the neural tube (Fig. 6A) and show an average length of  $104.1 \pm 13.47 \mu\text{m}$  (Fig. 6E). The treatment of wildtype embryos with  $0.08 \mu\text{g/ml}$  Lat. B from 20 to 28 hpf (Fig. 6B) resulted in a significant decrease in ISV length ( $61.53 \pm 23.27 \mu\text{m}$ ) compared to control embryos ( $p < 0.0001$ , Fig. 6E). A similar decrease in ISV length was also observed in *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos treated with DMSO ( $63.75 \pm 30.24 \mu\text{m}$ , Fig. 6C and E). When *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos were treated with  $0.08 \mu\text{g/ml}$  Lat. B from 20 to 28 hpf, there was a greater decrease in the length of ISV formed at 28 hpf ( $43.10 \pm 24.42 \mu\text{m}$ , Fig. 6D and E) when compared to the inhibition of actin polymerization ( $p = 0.0028$ ) or depletion of water influx ( $p < 0.0001$ ) alone (Movie S7). These results therefore demonstrate additive functions of actin polymerization and water influx in driving EC migration during sprouting angiogenesis (Fig. 6F).

## Discussion

In this study, we demonstrate that endothelial tip cells utilize two mechanisms of migration concurrently during sprouting angiogenesis in the zebrafish - actin polymerization and hydrostatic pressure. The increase in hydrostatic pressure in endothelial tip cells is generated by Aqp1a.1- and Aqp8a.1-mediated water inflow. By performing detailed expression analyses, we showed that *aqp1a.1* and *aqp8a.1* are differentially expressed in newly formed vascular sprouts. *Aqp1a.1* is expressed earlier in the DA than *aqp8a.1* during development and in ECs destined to be tip cells. In newly formed ISVs, endothelial tip cells express higher *aqp1a.1* mRNA levels while stalk cells express higher *aqp8a.1* mRNA levels. In zebrafish harbouring mutations in both *aqp1a.1* and *aqp8a.1*, there is delayed tip cell protrusion from the DA, decreased tip cell migration and increased occurrence of truncated ISVs at 3 dpf. We further discovered that *aqp1a.1* and *aqp8a.1* expression is upregulated by VEGFR2 activity, proposing that VEGFA-VEGFR2 signalling employs Aquaporin function to drive sprouting angiogenesis.

Several lines of evidence implicate Aquaporins in facilitating water flow into endothelial tip cells during sprouting angiogenesis. First, timelapse imaging showed enriched localization of Aqp1a.1 and Aqp8a.1 proteins at the migrating front of tip cells, suggesting that water flow occurs locally at the leading edge of the cell. Secondly, single cell analyses revealed Aqp1a.1- and Aqp8a.1-deficient tip cells have decreased cell volume, impaired expansion of cell membranes at the leading edge



**Figure 6.**

### Additive function of actin polymerization and hydrostatic pressure in driving EC migration and sprouting angiogenesis.

(A-D) Representative maximum intensity projection confocal z-stacks of 28 hpf wild type (A and B) and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* (C and D) embryos treated with 0.1% DMSO (A and C) or 0.8 μg/ml Lat. B (B and D) from 20 to 28 hpf (for each condition: wild type, n = 10 embryos; *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>*, n = 12 embryos, 2 independent experiments). Scale bar, 20 μm. (E) Quantification of ISV length in 28 hpf wild type and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos treated with 0.1% DMSO or 0.8 μg/ml Lat. B (wild type: n = 60 vessels from 10 control embryos, n = 66 vessels from 10 Lat. B-treated embryos; *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>*: n = 126 vessels from 12 control embryos, n = 112 vessels from 11 Lat. B-treated embryos, 2 independent experiments). Statistical significance was determined by Brown-Forsythe ANOVA test with Sidak's multiple comparison test; ns,  $p > 0.05$ , \*\* $p < 0.01$ , and \*\*\*\* $p < 0.0001$ . (F) Endothelial tip cells generate membrane protrusions by actin polymerization and increased hydrostatic pressure via Aquaporin-mediated water inflow. In the absence of actin polymerization and presence of Aquaporins, hydrostatic pressure deforms membranes to generate membrane protrusions. When Aquaporin function is lost, membrane protrusions is generated by actin polymerization. Fewer membrane protrusions are formed when only one mechanism is utilized, resulting in slower EC migration. See also Movie S7.

and decreased generation of stable cell protrusions. These observations implicate water flow into tip cells, leading to increased cell volume. Due to the incompressible property of water within an enclosed compartment such as the cell, water influx will lead to a rise in hydrostatic pressure. At the leading edge of tip cells, increased hydrostatic pressure would expand and deform membranes to generate protrusions. Additionally, water influx can reduce cytoplasmic viscosity and increase spacing between the plasma membrane and the underlying actin cytoskeleton, promoting actin monomer diffusion, actin polymerization and the generation of membrane protrusions to facilitate cell migration (Boer et al., 2023 [↗](#); Loitto et al., 2009 [↗](#)).

A key finding in this study is the demonstration that endothelial tip cells employ hydrostatic pressure as a second mechanism of cell migration *in vivo*. In a previous study, we discovered that ECs continue to migrate when actin polymerization is inhibited and in the absence of filopodia (Phng et al., 2013 [↗](#)). Here, we show that hydrostatic pressure is the driving force for EC migration in the absence of actin-based force generation. When both mechanisms are perturbed, there is a greater impairment in EC migration (**Fig. 6D and E** [↗](#)). This finding highlights the importance of hydrostatic pressure as an additional mechanism that reinforces actin-based cell shape changes and motility to ensure robust migration and formation of new blood vessels in complex three-dimensional environments. Such dual mode migration is advantageous when blood vessels develop in physically confined spaces such as in the zebrafish trunk, where ECs must emerge from the DA and migrate within somite boundaries and over tissues such as the notochord and neural tube to form ISVs. Such dependence on water flow in cell migration under confined microenvironment has been reported in human MDA-MB-231 breast cancer cells and mouse S180 sarcoma cells, which show cell migration persistence in narrow channels after inhibition of actin polymerization or myosin II-mediated contractility (Stroka et al., 2014 [↗](#)). In this context, AQP5 mediates water flow and the decrease or increase in AQP5 expression suppresses or enhances, respectively, cancer cell migration (Chae et al., 2008 [↗](#); Jung et al., 2011 [↗](#); Stroka et al., 2014 [↗](#)).

Our findings on the endothelial function of zebrafish Aqp1a.1, which is homologous to mammalian AQP1, corroborate previous studies demonstrating a function of AQP1 in promoting cell migration. In tumour bearing AQP1-null mice, there is reduced tumour angiogenesis and AQP1-deficient ECs display reduced migration in an *in vitro* wound healing assay (Maltaner et al., 2020 [↗](#); Saadoun et al., 2005 [↗](#)). In chick, AQP1 expression levels affect neural crest cell migration speed and direction by regulating filopodia length and stability (McLennan et al., 2019 [↗](#)). The regulation of membrane protrusions has also been demonstrated for other Aquaporin proteins. For example, AQP9 weakens membrane-cytoskeleton anchorage and promote formation of membrane protrusions such as filopodia and blebs (Karlsson et al., 2013 [↗](#); Loitto et al., 2007 [↗](#)). We also demonstrate for the first time a role of Aqp8a.1 in promoting EC migration and sprouting angiogenesis in the zebrafish. However, its mammalian ortholog, AQP8, was shown to inhibit migration of colorectal cancer cell lines *in vitro* (Wu et al., 2017 [↗](#)), suggesting cell- and context-dependent function of AQP8 in the regulation of cell migration.

A central determinant of water movement is the osmotic gradient across the membrane, with water flowing in the direction of higher solute concentration. Given a crucial role of water flow in regulating cell volume, shape and migration, it is also important to understand the function and distribution of ion channels, exchangers and transporters and water channels that generate an osmotic gradient across the cell. This is reflected in a growing number of studies demonstrating a role of ion transporters in cell migration. In neutrophils, the increase in cell volume and potentiation of cell migration depend on the sodium-proton exchanger 1 (NHE1) and the chloride-bicarbonate exchanger 2 (AE2) (Nagy et al., 2023 [↗](#)). In T cells, chemokine-induced migration depends on ion influx via SLC12A1, and water influx via AQP3 (Boer et al., 2023 [↗](#)). In tumour cells migrating in narrow channels, the establishment of a polarized distribution of  $\text{Na}^+/\text{H}^+$  pumps and Aquaporin proteins in cell membranes is required for net inflow of ions and water at the leading edge and net outflow at the trailing edge (Stroka et al., 2014 [↗](#)). We have also observed enrichment of Aqp1a.1 and Aqp8a.1 proteins in ECs, at the leading edge of migrating endothelial

tip cells as well as apical membranes during lumen formation (not shown). This leads to the question of how Aquaporin localisation is regulated as this will determine the site of increased water flow and hydrostatic pressure. The rapid changes in subcellular localization of mammalian Aquaporins upon stimulation occurs mainly via trafficking to and from the plasma membrane (Markou et al., 2022 [\[link\]](#)), and can lead to changes in the amount of Aquaporin protein in the plasma membrane. The regulation of AQP subcellular relocalization via calmodulin- and/or phosphorylation-dependent mechanisms has been implicated for AQP0-5 and AQP7-9 (Markou et al., 2022 [\[link\]](#)). For example, rapid translocation of AQP1 to plasma membrane upon a hypotonic stimulus is dependent on calmodulin activation and phosphorylation by protein kinases C (PKC) (Conner et al., 2012 [\[link\]](#)). Phosphorylation of AQP2 by protein kinase A (PKA) is required for vasopressin-stimulated relocalization of AQP2 from storage vesicles to the plasma membrane (Noda and Sasaki, 2005 [\[link\]](#)). Subcellular localization of AQP8 in hepatocytes is also regulated by PKA and PI3K signalling (Gradilone et al., 2005 [\[link\]](#), 2003 [\[link\]](#)). In mouse neutrophils, exposure to a chemotactic gradient caused translocation of AQP9 to the leading edge of the cell that is mediated by PKC-dependent phosphorylation (Karlsson et al., 2011 [\[link\]](#)). Further work is needed to elucidate the mechanism of Aqp1a.1 and Aqp8a.1 protein distribution in ECs and which ion transporters are expressed in ECs to regulate the ion movement and cell migration.

In this study, we have assumed that the function of Aqp1a.1 and Aqp8a.1 is a result of water permeation based on previous studies demonstrating water molecules flowing through mammalian AQP1 (Preston et al., 1992 [\[link\]](#); Zeidel et al., 1992 [\[link\]](#)) and AQP8 (Liu et al., 2006 [\[link\]](#); Ma et al., 1997 [\[link\]](#)) as well as through their zebrafish orthologs Aqp1a.1 and Aqp8a.1 (Tingaud-Sequeira et al., 2010 [\[link\]](#)). However, AQP1 and AQP8 are not water-specific channels. AQP1 can conduct unpolar gases carbon dioxide (Nakhoul et al., 1998 [\[link\]](#); Prasad et al., 1998 [\[link\]](#); Wang et al., 2007 [\[link\]](#)) and nitric oxide (Herrera et al., 2006 [\[link\]](#); Herrera and Garvin, 2007 [\[link\]](#)), and in zebrafish, Aqp1a.1 also mediates the transfer for small gaseous molecules such as carbon dioxide and ammonia (Talbot et al., 2015 [\[link\]](#)). AQP8 is permeable to ammonia (Jahn et al., 2004 [\[link\]](#); Liu et al., 2006 [\[link\]](#)) and is present in the inner mitochondrial membrane of different cells (Calamita et al., 2005 [\[link\]](#)), where it is suggested to mediate ammonia transport rather than water fluxes (Soria et al., 2010 [\[link\]](#)). Among the three Aqp8 paralogs in teleost, Aqp8a.1 and Aqp8a.2 are also permeable to urea (Tingaud-Sequeira et al., 2010 [\[link\]](#)) and Aqp8b was shown to act mainly as a mitochondrial H<sub>2</sub>O<sub>2</sub> channel (peroxiporin) in activated marine and freshwater teleost spermatozoa (Chauvigné et al., 2021 [\[link\]](#), 2015 [\[link\]](#)). Nevertheless, because water movement flows up an osmotic gradient, increased entry of ions into the cell through Aquaporins will also trigger water inflow.

In summary, our study highlights the role of water influx and hydrostatic pressure as another force-generating mechanism utilized by cells to build tissues during animal development. We demonstrate that endothelial tip cells employ both actin polymerization and hydrostatic pressure for robust sprouting angiogenesis. As morphogenetic events are governed by a coordination of cell shape changes, cell migration and rearrangements, we envision that the shaping and formation of other tissues will also depend on Aquaporin function and changes in hydrostatic pressure.

## Methods

### Zebrafish maintenance and stocks

Zebrafish (*Danio rerio*) were raised and staged according to established protocols (Kimmel et al., 1995 [\[link\]](#)). Transgenic lines used in this work are *Tg(kdrl:EGFP)<sup>s843</sup>* (Jin et al., 2005 [\[link\]](#)), *Tg(fli1a:H2B-EGFP)<sup>ncv69</sup>* [re-named in ZFIN as *Tg(fli1:zgc:114046-EGFP)*] (Ando et al., 2019 [\[link\]](#)), *Tg(fli1:Lifeact-mCherry)<sup>ncv7</sup>* (Wakayama et al., 2015 [\[link\]](#)), *Tg(fli1:myr-mCherry)<sup>ncv1</sup>* (Fukuhara et al., 2014 [\[link\]](#)), *Tg(fli1ep:EGFP-PLC1δPH)<sup>rk26</sup>* (Kondrychyn et al., 2020 [\[link\]](#)), *Tg(fli1:h2bc1-mCherry)<sup>ncv31</sup>* (Nakajima et al., 2017 [\[link\]](#)), *Tg(kdrl:ras-mCherry)<sup>s916</sup>* (Hogan et al., 2009 [\[link\]](#)). Zebrafish were maintained on a 12-hour light/12-hour dark cycle, and fertilized eggs were collected and raised in E3 medium at 28°C.

To inhibit pigmentation in embryos older than 24 hpf, 0.003% *N*-Phenylthiourea (Sigma-Aldrich) in E3 medium was used. All animal experiments were approved by the Institutional Animal Care and Use Committee at RIKEN Kobe Branch (IACUC).

## Plasmids and oligonucleotides

All DNA constructs used in this study were generated by using In-Fusion HD Cloning kit (Takara Bio Inc.). Plasmid *mEmerald-N1* was a gift from Michael Davidson (Addgene plasmid #53976). Plasmids *pDsChLexOG* (Emelyanov and Parinov, 2008 [DOI](#)) containing terminal *cis*-required sequences of the *Ds* transposon from maize was a generous gift from Sergei Parinov. A detailed information regarding plasmids and primers used in this study can be found in Supplementary Tables 1 and 2, respectively.

## Cloning of *aqp1a.1* and *aqp8a.1* genes

Total RNA was isolated from 1-day old zebrafish embryos with TRI Reagent using Direct-zol<sup>TM</sup> RNA MicroPrep kit (Zymo Research) according to the manufacturer's protocol. The first-strand cDNA was synthesized from 1 µg of a total RNA by oligo(dT) priming using the SuperScript III First-Strand synthesis system (Invitrogen) according to the manufacturer's protocol. Amplification of cDNAs was performed using a high-fidelity KOD-Plus-Neo DNA polymerase (Toyobo, Japan) and resulting PCR products were cloned using NEB<sup>®</sup> PCR Cloning kit (New England BioLabs). Positive clones and plasmids were verified by DNA sequencing.

## Mosaic expression of DNA constructs and transgenic lines generation

To establish *Tg(fli1ep:aqp1a.1-mEmerald)<sup>rk30</sup>* and *Tg(fli1ep:aqp8a.1-mEmerald)<sup>rk31</sup>* transgenic lines, *Tol2*-based plasmids (10 ng/µl) were co-injected into one-cell stage zebrafish embryos with *Tol2* transposase mRNA (100 ng/µl) transcribed from *NotI*-linearized *pCS-TP* plasmid (a gift from Koichi Kawakami, National Institute of Genetics, Japan) using the mMESSAGE mMACHINE SP6 kit (Invitrogen). Injected embryos were raised to adults and screened for germline transmission to establish a stable transgenic line. For mosaic expression of DNA construct, *Ds*-based *kdrl:mEmerald* plasmid (10 ng/µl) was co-injected into one-cell stage zebrafish embryos with *Ac* transposase mRNA (100 ng/µl) transcribed from *BamHI*-linearized *pAc-SP6* plasmid (Emelyanov et al., 2006 [DOI](#)) using the mMESSAGE mMACHINE SP6 kit (Invitrogen). Embryos were analyzed at 25–26 hpf after injections.

## In vivo cell volume analysis

Single-cell labelling of ECs in ISVs was achieved by mosaic expression of *pDs-kdrl:mEmerald* plasmid in wild type or *aqp1a.1<sup>rk28/rk28</sup>*, *aqp8a.1<sup>rk29/rk29</sup>* mutant *Tg(fli1:Lifeact-mCherry)<sup>ncv7</sup>* transgenic embryos. Embryos were imaged at 25–26 hpf with an Olympus UPLSAPO 60x/NA 1.2 water immersion objective with optical Z planes interval of 0.26 µm. Cell volume was measured with Huygens Essential 22.10 software (Scientific Volume Imaging B.V.) using object analyzer tool with manual adjustment of threshold.

## Generation of zebrafish *aqp1a.1* and *aqp8a.1* mutants

Aquaporins mutants were generated by CRISPR/Cas9-mediated mutagenesis. A guide RNAs (gRNAs) targeting the first exons of *aqp1a.1* and *aqp8a.1* (Fig. S4 and S5) were generated by using cloning-independent protocol (Gagnon et al., 2014 [DOI](#)). Briefly, to generate templates for gRNA transcription, a 60-base oligodeoxynucleotides containing the SP6 promoter sequence, the gene-specific sequence (*aqp1a.1*: 5'-GACAGCTGGCCAGCAGACCC-3'; *aqp8a.1*: 5'-GATGTCTCCCCATCGCCCG-3') and 23-base overlap region was annealed to an 80-base constant oligodeoxynucleotide encoding the reverse-complement of the tracrRNA tail. The ssDNA overhangs were filled-in with T4 DNA polymerase (New England BioLabs) to generate dsDNA. The

gRNAs were *in vitro* transcribed using MEGAScript SP6 kit (Invitrogen), DNase treated and purified with RNA Clean and Concentrator kit (Zymo Research). The gRNAs were quality-checked by running 1 µg of the product on a 2% TBE-agarose gel. Cas9/gRNA RNP complex was assembled just prior to injection and after 5 min incubation at room temperature 200 pg of Cas9 protein (Invitrogen) and 200 pg of gRNAs (100 pg of each, *aqp1a.1* gRNA and *aqp8a.1* gRNA) were co-injected into one-cell stage *Tg(fli1:myr-mCherry)<sup>ncv1</sup>* embryos. Such dose produced over 70% embryos survival post-injection showed CRISPR/Cas9-induced somatic *aqp1a.1* and *aqp8a.1* gene mutations. F<sub>0</sub> founders were identified by outcrossing CRISPR/Cas9-injected fish with wild type fish and screening the offspring for mutations at 2 dpf using Sanger sequencing. We found one F<sub>0</sub> fish with germline transmitted mutations in both *aqp1a.1* and *aqp8a.1* genes and outcrossed it with *Tg(fli1:myr-mCherry)<sup>ncv1</sup>* and *Tg(fli1a:H2B-EGFP)<sup>ncv69</sup>* transgenic fish to establish F<sub>1</sub> generation of heterozygotes. A 55 F<sub>1</sub> fish were genotyped to identify 4 *aqp1a.1<sup>+rk28</sup>*, 11 *aqp8a.1<sup>+rk29</sup>* and 12 *aqp1a.1<sup>+rk28</sup>,aqp8a.1<sup>+rk29</sup>* heterozygote fish. Subsequently, F<sub>1</sub> heterozygote fish were in-crossed to establish a single and double homozygote fish.

## DNA isolation and genotyping

Genomic DNA was isolated from either embryos or fin clips using HotSHOT method (Meeker et al., 2007 [\[1\]](#)). To identify genomic lesions, the following primers were used: *aqp1a1*-fwd (5'-CGCTCCAGATTCATTAGCAGGA-3') and *aqp1a1*-rev (5'-GTAAGTGAAGTCTGCCAGTGA-3') to amplify a 550 bp fragment, and *aqp8a1*-fwd (5'-GGATCAATTGAGTTGCATAACAGAC-3') and *aqp8a1*-rev (5'-CTGTAATGTAGACTGTAAAGTGGA-3') to amplify a 638 bp fragment. Mutations were assessed by direct sequencing of purified PCR products.

## Quantitative real-time PCR

Total RNA from whole embryos was isolated with TRI reagent using Direct-zol RNA MicroPrep kit (Zymo Research) according to the manufacturer's protocol, including on-column DNA digestion. RNA was quantified using a NanoDrop 1000 spectrophotometer (ThermoFisher Scientific, USA). cDNA was synthesized from 300 ng of purified RNA using LunaScript RT SuperMix kit (New England BioLabs) according to the manufacturer's protocol. Amplification of target cDNA was performed in technical triplicate of three biological replicates using the SYBR green methods. Each qPCR reaction mixture contained 5 µL 2x Luna Universal qPCR master mix (New England BioLabs), 1 µL cDNA (2-fold dilution), 0.2 µL antarctic thermolabile UDG (New England BioLabs) and 250 nM each primer to a final volume of 10 µL. Reactions were run in 384-well plates (Applied Biosystems) using the QuantStudio 5 Real-Time PCR system (Applied Biosystems) with the following thermal cycling conditions: initial UDG treatment at 25°C for 2 minutes, UDG inactivation at 50°C for 5 min and denaturation at 95°C for 1 minutes, followed by 45 cycles of 15 sec at 95°C, 30 sec at 60°C with a plate read at the end of the extension step. Control reactions included a no-template control (NTC) and a no-reverse transcriptase control (NRT). Dissociation analysis of the PCR products was performed by running a gradient from 60 to 95°C to confirm the presence of a single PCR product. Amplification data was analyzed using QuantStudio Design and Analysis software v1.5.3 (Applied Biosystems) and relative fold change was calculated using Pfaffl method (Pfaffl, 2001 [\[2\]](#)) and normalized to *gapdh* expression (as an internal control). Primers are listed in Supplementary Table 2. Two embryos were used in each biological replicate to generate an average value that was used to calculate the final mean ± SD from two or three independent experiments.

## RNA *in situ* hybridization

## RNAscope *in situ* hybridization

RNAscope® *in situ* hybridization was conducted by using RNAscope Multiplex Fluorescent Reagent kit v2 (Advanced Cell Diagnostics). We adapted manufacture's protocol designed for samples mounted on slides and protocol developed for whole-mount embryo samples (Gross-Thebing et al., 2014). For each experimental point, 5 embryos were processed in one 1.5 ml Eppendorf tube. Briefly, 20 and 22 hpf old embryos were manually dechorionated and fixed in freshly prepared 4% methanol-free PFA in PBS for 1 hour at room temperature. After fixation embryos were washed three times in PBS containing 0.01% Tween-20 (PBST), dehydrated stepwise in 5-min washes in a series of increasing methanol concentrations (25%, 50%, 75%, 100%) in PBS and then stored in 100% methanol at -20°C before use them for hybridization. The methanol-stored embryos were incubated with 5% H<sub>2</sub>O<sub>2</sub> in methanol for 20 min at room temperature and then rehydrated stepwise in 5-min washes in series of decreasing methanol concentrations (75%, 50%, 25%) in PBS, followed by three times 5-min washes in PBST. Embryos were permeabilized in 2 drops of RNAscope® Protease III for 15 min, rinsed three times with PBST and hybridized with RNAscope target probes overnight at 40°C in water bath. After hybridization probes were recovered and embryos were washed three times with 0.2x SSCT [0.01% Tween-20 in SSC] at room temperature, re-fixed with 4% PFA for 10 min, and washed three times with 0.2x SSCT. For RNA detection, the embryos were first hybridized with three different amplifier solution for 30 min (Amp1, and Amp2) and 15 min (Amp3) in a water bath at 40°C. After each hybridization step, the embryos were washed three times with 0.2x SSCT for 15 min at room temperature. To develop signal of each probe, the embryos were sequentially incubated in a water bath at 40°C with: (i) the horseradish peroxidase (HRP) for 15 min, (ii) TSA fluorophore for 30 min and (iii) the HRP blocker for 15 min. After each incubation step, the embryos were washed three times with 0.2x SSCT for 15 min at room temperature. HRP step is linked to a probe channel, we first developed C1 probe, *Dr-aqp1a.1* (Advanced Cell Diagnostics) using HRP-C1 and TSA Vivid Fluorophore 520 (Tocris, dilution 1:1500), and then C3 probe, *Dr-aqp8a.1* (Advanced Cell Diagnostics) using HRP-C3 and TSA Vivid Fluorophore 650 (Tocris, dilution 1:1500). Nuclei were counterstained with DAPI ready-to-use solution overnight at 4°C. Prior to imaging embryos were rinsed in PBST and kept in 70% glycerol in PBS. Images were acquired using Olympus FV3000 confocal microscope and an Olympus UPlanXApo 60x/NA 1.42 oil immersion objective.

## Chromogenic *in situ* hybridization

Chromogenic *in situ* hybridization was conducted according to standard protocol (Thisse and Thisse, 2007) with minor modifications. For each experimental point, 10 embryos were processed in one 1.5 ml Eppendorf tube. Briefly, 30, 48, 72 and 96 hpf old embryos were manually dechorionated and fixed in freshly prepared 4% methanol-free PFA in PBS (pH 7.4) at 4°C overnight. After fixation embryos were washed three times in PBS containing 0.1% Tween-20 (PBST), dehydrated stepwise in 5-min washes in a series of increasing methanol concentrations (25%, 50%, 75%, 100%) in PBS and then stored in 100% methanol at -20°C before use them for hybridization. The methanol-stored embryos were rehydrated stepwise in 5-min washes in series of decreasing methanol concentrations (75%, 50%, 25%) in PBS followed by three times 5-min washes in PBST, and permeabilized 20 min with 10 µg/mL proteinase K (ThermoFisher Scientific). Hybridization was carried out in buffer [50% formamide, 5x SSC, 50 g/mL heparin, 500 g/mL tRNA (Roche) and 0.1% Tween-20] containing 5% dextran sulfate (MW=500 kDa) at 69°C overnight. After stringency wash, the specimens were blocked with 2% blocking reagent (Roche) in maleic acid buffer [100 mM maleic acid, 150 mM NaCl, 50 mM MgCl<sub>2</sub> and 0.1% Tween-20, pH 7.5] and incubated overnight with anti-digoxigenin antibody, conjugated with alkaline phosphatase (dilution 1:5000, Roche) at 4°C. For detection of alkaline phosphatase, specimens were incubated in staining buffer [50 mM Tris-HCl, 50 mM NaCl, 25 mM MgCl<sub>2</sub>, 2% Polyvinyl alcohol and 0.1% Tween-20, pH 9.5] containing 375 µg/mL nitro-blue tetrazolium chloride (Roche) and 175 µg/mL 5-bromo-4-chloro-3'-indolyl-phosphate (Roche). Embryos were cleared in 70% glycerol overnight and imaged using Leica M205FA microscope. **RNA probe synthesis.** The cDNA-containing vectors were

linearized with appropriate restriction enzymes (detailed maps of vectors will be provided upon request) and used as a template for RNA probe synthesis. Sense and antisense RNA probes were synthesized using MEGAscript SP6 or T7 kits (Invitrogen) and digoxigenin (DIG)-labeled rNTPs (Roche) according to the manufacturer's protocol and purified using RNA Clean and Concentrator kit (Zymo Research). The following sense and antisense DIG-labeled riboprobes were generated: (i) *aqp1a.1*, 1.1 kb long probe comprised 121 nt of 5'UTR, 783 nt of the open reading frame (ORF) and 207 nt of 3'UTR; (ii) *aqp8a.1*, 0.8 kb long probe comprised 17 nt of 5'UTR, 783 nt of ORF and 30 nt of 3'UTR. Sense probes showed no specific or unspecific staining (data not shown).

## Image processing

Chromogenic and RNAscope *in situ* hybridization images were processed using ImageJ software (NIH, version 2.9.0/1.53t) with brightness and contrast adjustments. Z-stacks of fluorescent images are presented as maximum intensity projections.

## Analysis of mRNA expression in tip and stalk cells

Embryos were mounted on slide and imaged using Olympus FV3000 confocal microscope and Olympus UPlanXApo 60x/NA 1.42 oil immersion objective. Maximum intensity projection of z-stacks was used to determine the level of cellular fluorescence in ImageJ. The tip and stalk cells were selected using freehand selection tool, "mean grey value", "area" and "integrated density" in each ROI were measured first for channel 1 (*aqp1a.1*). A region next to cell that has no fluorescence was selected as background and measured. The step was repeated for channel 2 (*aqp8a.1*) on the same ROIs. Two parameters were calculated for each channel, (i) the corrected total cell fluorescence,  $CTCF = \text{Integrated Density} - (\text{Area of selected cell} \times \text{Mean fluorescence of background})$  and (ii) ratio,  $R = CTCF \text{ of tip cell} / CTCF \text{ of stalk cell}$ . We consider that if  $R < 0.9$ , expression is lower in tip cell, if  $R = 0.9 - 1.1$ , expression is equal in tip and stalk cells, if  $R > 1.1$ , expression is higher in tip cell.

## Imaging

For live confocal imaging, embryos were mounted in 0.8% low-melt agarose (Bio-Rad) in E3 medium containing 0.16 mg/mL Tricaine (Sigma-Aldrich) and 0.003% phenylthiourea (Sigma-Aldrich) in glass bottom 35-mm dishes (MatTek). Confocal Z-stacks were acquired using an inverted Olympus IX83/Yokogawa CSU-W1 spinning disc confocal microscope equipped with a Zyla 4.2 CMOS camera (Andor) and Olympus UPLSAPO 40x/NA 1.25 or 30x/NA 1.05 silicone oil immersion objectives. Bright-field images were acquired on Leica M205FA microscope. Images were processed using ImageJ software (NIH, version 2.9.0/1.53t).

## Analysis of cell and nuclei migration

*aqp1a.1<sup>+/rk28</sup>;aqp8a.1<sup>+/rk29</sup>* and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos on *Tg(fli1a:H2B-EGFP)<sup>ncv69</sup>;fli1:Lifeact-mCherry)<sup>ncv7</sup>* double transgenic background were imaged from 21 to 30 hpf with time interval 6 min using Olympus UPLSAPO 30x/NA 1.05 silicon oil immersion objective. We tried to mount both heterozygote and mutant embryos on the same dish (1 het plus 2 mut, or 2 het plus 2 mut) to image embryos on the same experimental conditions. Acquired time-lapse images were registered using HyperStackReg plugin in ImageJ, tip cell and nuclei were tracked using Manual Tracking plugin and velocity was calculated.

## Analysis of grow rate of actin bundles in filopodia

Wild type and *aqp1a.1<sup>rk28/rk28</sup>;aqp8a.1<sup>rk29/rk29</sup>* embryos on *Tg(fli1ep:EGFP-PLC18PH)<sup>rk26</sup>;fli1:Lifeact-mCherry)<sup>ncv7</sup>* double transgenic background were imaged starting from 25 hpf for 10 min with time interval 30 sec using Olympus UPLSAPO 60x/NA 1.2 water immersion objective. Actin bundles growth was tracked using Manual Tracking plugin and velocity was calculated.

## Analysis of filopodia number and length

Wild type, *aqp1a.1*<sup>+/rk28</sup>; *aqp8a.1*<sup>+/rk29</sup> and *aqp1a.1*<sup>rk28/rk28</sup>; *aqp8a.1*<sup>rk29/rk29</sup> embryos on *Tg(fli1ep:EGFP-PLC1δPH)*<sup>rk26</sup>; *(fli1:Lifeact-mCherry)*<sup>ncv7</sup> double transgenic background were imaged starting from 24 hpf for 30 min with time interval of 50 sec using Olympus UPLSAPO 60x/NA 1.2 water immersion objective. In a single image we counted filopodia number per vessel, measured membrane length and calculated filopodia number.

## Chemical treatment

Latrunculin B (Merck Millipore) was dissolved in DMSO to 1 mg/ml and stored at -20°C. Dibenazepine (YO-01027) (Selleck Chemicals) was prepared as 10 mM solution in DMSO and stored at -80°C. Ki8751 (Selleck Chemicals) was prepared as 5 mM solution in DMSO and stored at -80°C. All compounds were diluted to the desired concentration in E3 medium.

## Statistical analysis

Statistical analysis was performed using Prism software version 10.2.0 (GraphPad). The variance between the mean values of two groups was evaluated using the unpaired Student's *t*-test. For assessment of more than three groups, we used one-way analysis of variance (ANOVA) test. A *P* value of <0.05 was considered statistically significant. Statistic details can be found in each figure legend.

## Cell culture

Adult human aortic endothelial cells (HAEC) (Lonza, lot-20TL231227) were cultured in EGM medium (Lonza) and used at passage 3. Cells were seeded in EBM medium (Lonzo) supplemented with 2% FBS (Gibco) at  $4 \times 10^5$  cells/well on 24-well plate (Trueline) coated with 5 µg/ml fibronectin (Sigma-Aldrich). Cells were treated with either 0.01% DMSO or different concentrations of Ki8751 inhibitor for 6 hours. Cells were lysed with TRI reagent and RNA was isolated using Direct-zol RNA MicroPrep kit (Zymo Research).

## Single-cell RNA sequencing

ECs were isolated from *Tg(kdr1:EGFP)*<sup>s843</sup> transgenic embryos. Cell sorting was carried out with FACSARIAII Cell Sorter (BD Bioscience). Single-cell suspension was loaded into the 10X Chromium system and cDNA libraries were constructed using Chromium Next GEM Single Cell 3' GEM, Library and Gel Bead Kit v2 (10X Genomics) according to manufacturer's protocol. Library was sequenced on the Illumina HiSeq 1500 Sequencer (Illumina, USA). Cell Ranger v2.1 was used to demultiplex raw base call (BCL) files generated by Illumina sequencers into FASTQ files, perform the alignment, barcode counting, and UMI counting. Sequencing reads were mapped to the zebrafish genome assembly (GRCz11, Ensembl release 92). Further analyses were performed using Seurat package49 (version 4.3.0) in R software (<https://www.r-project.org>). The expression matrices were first filtered by keeping genes that are expressed in a minimum of 3 cells and cells that expressed a minimum of 200 genes for downstream analysis. The data were then normalized using NormalizeData function which normalizes the gene expression for each cell by the total expression counts. To correct for batch effect between the data from the two-time points, the rpca method in the Seurat package was applied to integrate the data. The top 2000 variable genes identified using the vst method in FindVariableFeatures function were used for principal component analysis in RunPCA function, and the first 30 principal components were used for visualization analysis with Uniform Manifold Approximation and Projection (UMAP) method, and in FindNeighbors function analysis. The cell clustering resolution was set at 0.5 in FindClusters function.

## Acknowledgements

We thank members of the Phng Lab, Y-C. Wang and S. Thukral for discussions and suggestions; G. Chen, E. Taniguchi, RIKEN BDR Research Aquarium and RIKEN Kobe BioImaging Facilities & Factory for technical assistance; J. Vermot, M. Francois and A. Yap for comments on the manuscript. This work was supported by core funding from RIKEN BDR (to L.K.P.), RIKEN BDR-Otsuka Pharmaceutical Collaboration Center (I.K.), the Naito Foundation (L.K.P.), the JSPS Grants-in-Aid for Scientific Research (KAKENHI) grants (22H02624, 22H05168 to L.K.P; 22K06244 to I.K.), Swedish Research Council (2015-00550 to C.B.); Swedish Cancer Society (2018/449, 2018/1154 to C.B.); Knut and Alice Wallenberg Foundation (2020.0057, 2018.0218 to C.B.); Swedish Brain Foundation (ALZ2019-0130, ALZ2022-0005 to C.B.); the Leducq Foundation (22CVD01, 23CVD02 to C.B.) and the Innovative Medicines Initiative (IM2PACT-807015 to C.B.)

## Authorship contributions

Conceptualization: L.K.P.; Methodology: I.K., H.W. and L.K.P.; Formal analysis: I.K. and L.H.; Resources: C.B. and L.K.P.; Writing: I.K. and L.K.P.; Supervision: L.K.P.

## Declaration of interests

The authors declare no competing interests.

## References

1. Agre P, King LS, Yasui M, Guggino WB, Ottersen OP, Fujiyoshi Y, Engel A, Nielsen S (2002) **Aquaporin water channels – from atomic structure to clinical medicine** *J Physiol* **542**:3–16 <https://doi.org/10.1113/jphysiol.2002.020818>
2. Ando K *et al.* (2019) **Peri-arterial specification of vascular mural cells from naïve mesenchyme requires Notch signaling** *Development* **146**:dev165589–13 <https://doi.org/10.1242/dev.165589>
3. Baeyens N *et al.* (2015) **Vascular remodeling is governed by a VEGFR3-dependent fluid shear stress set point** *Elife* **4** <https://doi.org/10.7554/elife.04645>
4. Benedito R, Hellström M (2013) **Notch as a hub for signaling in angiogenesis** *Exp Cell Res* :1–8
5. Benfenati V, Caprini M, Dovizio M, Mylonakou MN, Ferroni S, Ottersen OP, Amiry-Moghaddam M. (2011) **An aquaporin-4/transient receptor potential vanilloid 4 (AQP4/TRPV4) complex is essential for cell-volume control in astrocytes** *Proc Natl Acad Sci USA* **108**:2563–8 <https://doi.org/10.1073/pnas.1012867108>
6. Bill BR, Korzh V (2014) **Choroid plexus in developmental and evolutionary perspective** *Front Neurosci* **8** <https://doi.org/10.3389/fnins.2014.00363>
7. Boer LL de *et al.* (2023) **T cell migration requires ion and water influx to regulate actin polymerization** *Nat Commun* **14** <https://doi.org/10.1038/s41467-023-43423-8>
8. Bussmann J, Wolfe SA, Siekmann AF (2011) **Arterial-venous network formation during brain vascularization involves hemodynamic regulation of chemokine signaling** *Development* **138**:1717–1726
9. Calamita G, Ferri D, Gena P, Liquori GE, Cavalier A, Thomas D, Svelto M (2005) **The Inner Mitochondrial Membrane Has Aquaporin-8 Water Channels and Is Highly Permeable to Water** *J Biol Chem* **280**:17149–17153 <https://doi.org/10.1074/jbc.c400595200>
10. Chae YK *et al.* (2008) **Expression of Aquaporin 5 (AQP5) Promotes Tumor Invasion in Human Non Small Cell Lung Cancer** *PLoS ONE* **3** <https://doi.org/10.1371/journal.pone.0002162>
11. Chan CJ, Costanzo M, Ruiz-Herrero T, Mönke G, Petrie RJ, Bergert M, Diz-Muñoz A, Mahadevan L, Hiiragi T (2019) **Hydraulic control of mammalian embryo size and cell fate** *Nature* **571**:112–116 <https://doi.org/10.1038/s41586-019-1309-x>
12. Chang Y-F, Imam JS, Wilkinson MF (2007) **The Nonsense-Mediated Decay RNA Surveillance Pathway** *Annu Rev Biochem* **76**:51–74 <https://doi.org/10.1146/annurev.biochem.76.050106.093909>
13. Chauvigné F, Boj M, Finn RN, Cerdà J (2015) **Mitochondrial aquaporin-8-mediated hydrogen peroxide transport is essential for teleost spermatozoon motility** *Sci Rep* **5** <https://doi.org/10.1038/srep07789>

14. Chauvigné F, Ducat C, Ferré A, Hansen T, Carrascal M, Abián J, Finn RN, Cerdà J (2021) **A multiplier peroxiporin signal transduction pathway powers piscine spermatozoa** *Proc Natl Acad Sci USA* **118** <https://doi.org/10.1073/pnas.2019346118>
15. Choudhury MI, Benson MA, Sun SX (2022) **Trans-epithelial fluid flow and mechanics of epithelial morphogenesis** *Semin Cell Dev Biol* **131**:146–159 <https://doi.org/10.1016/j.semcdb.2022.05.020>
16. Chugh M, Munjal A, Megason SG (2022) **Hydrostatic pressure as a driver of cell and tissue morphogenesis** *Semin Cell Dev Biol* **131**:134–145 <https://doi.org/10.1016/j.semcdb.2022.04.021>
17. Clarke DN, Martin AC (2021) **Actin-based force generation and cell adhesion in tissue morphogenesis** *Curr Biol* **31**:R667–R680 <https://doi.org/10.1016/j.cub.2021.03.031>
18. Conner MT, Conner AC, Bland CE, Taylor LHJ, Brown JEP, Parri HR, Bill RM (2012) **Rapid Aquaporin Translocation Regulates Cellular Water Flow** *J Biol Chem* **287**:11516–11525 <https://doi.org/10.1074/jbc.m111.329219>
19. Day RE, Kitchen P, Owen DS, Bland C, Marshall L, Conner AC, Bill RM, Conner MT (2014) **Human aquaporins: regulators of transcellular water flow** *Biochim Biophys Acta* **1840**:1492–1506 <https://doi.org/10.1016/j.bbagen.2013.09.033>
20. Dumortier JG, Verge-Serandour ML, Tortorelli AF, Mielke A, Plater L de, Turlier H, Maître J-L. (2019) **Hydraulic fracturing and active coarsening position the lumen of the mouse blastocyst** *Science* **365**:465–468 <https://doi.org/10.1126/science.aaw7709>
21. Emelyanov A, Gao Y, Naqvi NI, Parinov S (2006) **Trans-Kingdom Transposition of the Maize Dissociation Element** *Genetics* **174**:1095–1104 <https://doi.org/10.1534/genetics.106.061184>
22. Emelyanov A, Parinov S (2008) **Mifepristone-inducible LexPR system to drive and control gene expression in transgenic zebrafish** *Dev Biol* **320**:113–121 <https://doi.org/10.1016/j.ydbio.2008.04.042>
23. Farinas J, Kneen M, Moore M, Verkman AS (1997) **Plasma Membrane Water Permeability of Cultured Cells and Epithelia Measured by Light Microscopy with Spatial Filtering** *J Gen Physiol* **110**:283–296 <https://doi.org/10.1085/jgp.110.3.283>
24. Figueiredo AM *et al.* (2021) **Endothelial cell invasion is controlled by dactylopodia** *Proc Natl Acad Sci USA* **118** <https://doi.org/10.1073/pnas.2023829118>
25. Fukuhara S, Zhang J, Yuge S, Ando K, Wakayama Y, Sakaue-Sawano A, Miyawaki A, Mochizuki N (2014) **Visualizing the cell-cycle progression of endothelial cells in zebrafish** *Dev Biol* **393**:10–23 <https://doi.org/10.1016/j.ydbio.2014.06.015>
26. Gagnon JA *et al.* (2014) **Efficient mutagenesis by Cas9 protein-mediated oligonucleotide insertion and large-scale assessment of single-guide RNAs** *PLoS ONE* **9** <https://doi.org/10.1371/journal.pone.0098186>
27. Gebala V, Collins R, Geudens I, Phng L-K, Gerhardt H (2016) **Blood flow drives lumen formation by inverse membrane blebbing during angiogenesis in vivo** *Nat Cell Biol* **18**:443–450 <https://doi.org/10.1038/ncb3320>
28. Gerhardt H *et al.* (2003) **VEGF guides angiogenic sprouting utilizing endothelial tip cell filopodia** *J Cell Biol* **161** <https://doi.org/10.1083/jcb.200302047>

29. Gradilone SA, Carreras FI, Lehmann GL, Marinelli RA (2005) **Phosphoinositide 3-kinase is involved in the glucagon-induced translocation of aquaporin-8 to hepatocyte plasma membrane** *Biol Cell* **97**:831–836 <https://doi.org/10.1042/bc20040115>
30. Gradilone SA, García F, Huebert RC, Tietz PS, Larocca MC, Kierbel A, Carreras FI, Larusso NF, Marinelli RA (2003) **Glucagon induces the plasma membrane insertion of functional aquaporin-8 water channels in isolated rat hepatocytes** *Hepatology* **37**:1435–41 <https://doi.org/10.1053/jhep.2003.50241>
31. Gross-Thebing T, Paksa A, Raz E (2014) **Simultaneous high-resolution detection of multiple transcripts combined with localization of proteins in whole-mount embryos** *BMC Biol* **12** <https://doi.org/10.1186/s12915-014-0055-7>
32. Groth C, Alvord WG, Quiñones OA, Fortini ME (2010) **Pharmacological Analysis of *Drosophila melanogaster* γ-Secretase with Respect to Differential Proteolysis of Notch and APP** *Mol Pharmacol* **77**:567–574 <https://doi.org/10.1124/mol.109.062471>
33. Herrera M, Garvin JL (2007) **Novel role of AQP-1 in NO-dependent vasorelaxation** *Am J Physiol Ren Physiol* **292**:F1443–F1451 <https://doi.org/10.1152/ajprenal.00353.2006>
34. Herrera M, Hong NJ, Garvin JL (2006) **Aquaporin-1 Transports NO Across Cell Membranes** *Hypertension* **48**:157–164 <https://doi.org/10.1161/01.hyp.0000223652.29338.77>
35. Hogan BM, Bos FL, Bussmann J, Witte M, Chi NC, Duckers HJ, Schulte-Merker S (2009) **ccbe1 is required for embryonic lymphangiogenesis and venous sprouting** *Nat Genet* **41** <https://doi.org/10.1038/ng.321>
36. Hu J, Verkman AS, Hu J, Verkman AS (2006) **Increased migration and metastatic potential of tumor cells expressing aquaporin water channels** *FASEB J* **20**:1892–1894 <https://doi.org/10.1096/fj.06-5930fje>
37. Jahn TP, Møller ALB, Zeuthen T, Holm LM, Klærke DA, Mohsin B, Kühlbrandt W, Schjoerring JK (2004) **Aquaporin homologues in plants and mammals transport ammonia** *FEBS Lett* **574**:31–36 <https://doi.org/10.1016/j.febslet.2004.08.004>
38. Jin S-W, Beis D, Mitchell T, Chen J-N, Stainier DY (2005) **Cellular and molecular analyses of vascular tube and lumen formation in zebrafish** *Development* **132** <https://doi.org/10.1242/dev.02087>
39. Jung HJ, Park J-Y, Jeon H-S, Kwon T-H (2011) **Aquaporin-5: A Marker Protein for Proliferation and Migration of Human Breast Cancer Cells** *PLoS ONE* **6** <https://doi.org/10.1371/journal.pone.0028492>
40. Karlsson T, Bolshakova A, Magalhães MAO, Loitto VM, Magnusson K-E (2013) **Fluxes of Water through Aquaporin 9 Weaken Membrane-Cytoskeleton Anchorage and Promote Formation of Membrane Protrusions** *PLoS ONE* **8** <https://doi.org/10.1371/journal.pone.0059901>
41. Karlsson T, Glogauer M, Ellen RP, Loitto V, Magnusson K, Magalhães MAO (2011) **Aquaporin 9 phosphorylation mediates membrane localization and neutrophil polarization** *J Leukoc Biol* **90**:963–973 <https://doi.org/10.1189/jlb.0910540>
42. Kimmel CB, Ballard WW, Kimmel SR, Ullmann B, Schilling TF (1995) **Stages of embryonic development of the zebrafish** *Dev Dyn* **203**:253–310

43. Kondrychyn I, Kelly DJ, Carretero NT, Nomori A, Kato K, Chong J, Nakajima H, Okuda S, Mochizuki N, Phng L-K (2020) **Marcksl1 modulates endothelial cell mechanoreponse to haemodynamic forces to control blood vessel shape and size** *Nat Commun* **11**:5476–18 <https://doi.org/10.1038/s41467-020-19308-5>
44. Koun S, Kim J-D, Rhee M, Kim M-J, Huh T-L (2016) **Spatiotemporal expression pattern of the zebrafish aquaporin 8 family during early developmental stages** *Gene Expr Patterns* **21**:1–6 <https://doi.org/10.1016/j.gep.2016.06.001>
45. Kozono D, Yasui M, King LS, Agre P (2002) **Aquaporin water channels: atomic structure molecular dynamics meet clinical medicine** *J Clin Invest* **109**:1395–1399 <https://doi.org/10.1172/jci15851>
46. Li Y, Konstantopoulos K, Zhao R, Mori Y, Sun SX (2020) **The importance of water and hydraulic pressure in cell dynamics** *J Cell Sci* **133** <https://doi.org/10.1242/jcs.240341>
47. Liu K, Nagase H, Huang CG, Calamita G, Agre P (2006) **Purification and functional characterization of aquaporin-8** *Biol Cell* **98**:153–161 <https://doi.org/10.1042/bc20050026>
48. Loitto VM, Huang C, Sigal YJ, Jacobson K (2007) **Filopodia are induced by aquaporin-9 expression** *Exp Cell Res* **313**:1295–1306 <https://doi.org/10.1016/j.yexcr.2007.01.023>
49. Loitto VM, Karlsson T, Magnusson K (2009) **Water flux in cell motility: Expanding the mechanisms of membrane protrusion** *Cell Motil Cytoskeleton* **66**:237–247 <https://doi.org/10.1002/cm.20357>
50. Ma T, Yang B, Verkman AS (1997) **Cloning of a Novel Water and Urea-Permeable Aquaporin from Mouse Expressed Strongly in Colon, Placenta, Liver, and Heart** *Biochem Biophys Res Commun* **240**:324–328 <https://doi.org/10.1006/bbrc.1997.7664>
51. Mallavarapu A, Mitchison T (1999) **Regulated Actin Cytoskeleton Assembly at Filopodium Tips Controls Their Extension and Retraction** *J Cell Biol* **146**:1097–1106 <https://doi.org/10.1083/jcb.146.5.1097>
52. Maltaner RE, Schiappacasse A, Chamorro ME, Nesse AB, Vittori DC (2020) **Aquaporin-1 plays a key role in erythropoietin-induced endothelial cell migration** *Biochim Biophys Acta Mol Cell Res* **1867** <https://doi.org/10.1016/j.bbamcr.2019.118569>
53. Marikawa Y, Alarcon VB (2012) **Mouse Development, From Oocyte to Stem Cells** *Results Probl Cell Differ* **55**:165–184 [https://doi.org/10.1007/978-3-642-30406-4\\_9](https://doi.org/10.1007/978-3-642-30406-4_9)
54. Markou A *et al.* (2022) **Molecular mechanisms governing aquaporin relocalisation** *Biochim Biophys Acta Biomembr* **1864** <https://doi.org/10.1016/j.bbamem.2021.183853>
55. McLennan R *et al.* (2019) **Neural crest cells bulldoze through the microenvironment using Aquaporin 1 to stabilize filopodia** *Development* **147** <https://doi.org/10.1242/dev.185231>
56. Meeker ND, Hutchinson SA, Ho L, Trede NS. (2007) **Method for isolation of PCR-ready genomic DNA from zebrafish tissues** *Biotech* **43**:610–614

57. Mosaliganti KR, Swinburne IA, Chan CU, Obholzer ND, Green AA, Tanksale S, Mahadevan L, Megason SG (2019) **Size control of the inner ear via hydraulic feedback** *Elife* **8** <https://doi.org/10.7554/elife.39596>
58. Munjal A, Lecuit T (2014) **Actomyosin networks and tissue morphogenesis** *Development* **141**:1789–1793 <https://doi.org/10.1242/dev.091645>
59. Murrell M, Oakes PW, Lenz M, Gardel ML (2015) **Forcing cells into shape: the mechanics of actomyosin contractility** *Nat Cell Biol* **16**:486–498 <https://doi.org/10.1038/nrm4012>
60. Nagy TL, Strickland J, Weiner OD (2023) **Neutrophils actively swell to potentiate rapid migration** *bioRxiv* <https://doi.org/10.1101/2023.05.15.540704>
61. Nakajima H *et al.* (2017) **Flow-Dependent Endothelial YAP Regulation Contributes to Vessel Maintenance** *Dev Cell* **40**:523–536 <https://doi.org/10.1016/j.devcel.2017.02.019>
62. Nakhoul NL, Davis BA, Romero MF, Boron WF (1998) **Effect of expressing the water channel aquaporin-1 on the CO<sub>2</sub> permeability of *Xenopus* oocytes** *Am J Physiol Cell Physiol* **274**:C543–C548 <https://doi.org/10.1152/ajpcell.1998.274.2.c543>
63. Noda Y, Sasaki S (2005) **Trafficking mechanism of water channel aquaporin-2** *Biol Cell* **97**:885–892 <https://doi.org/10.1042/bc20040120>
64. Papadopoulos MC, Saadoun S, Verkman AS (2008) **Aquaporins and cell migration** *Pflugers Arch* **456**:693–700 <https://doi.org/10.1007/s00424-007-0357-5>
65. Parab S, Quick RE, Matsuoka RL (2021) **Endothelial cell-type-specific molecular requirements for angiogenesis drive fenestrated vessel development in the brain** *Elife* **10** <https://doi.org/10.7554/elife.64295>
66. Pfaffl MW (2001) **A new mathematical model for relative quantification in real-time RT-PCR** *Nucleic Acids Res* **29**:e45–e45 <https://doi.org/10.1093/nar/29.9.e45>
67. Phng LK, Gerhardt H (2009) **Angiogenesis: A Team Effort Coordinated by Notch** *Dev Cell* **16**:196–208 <https://doi.org/10.1016/j.devcel.2009.01.015>
68. Phng LK, Stanchi F, Gerhardt H (2013) **Filopodia are dispensable for endothelial tip cell guidance** *Development* **140**:4031–4040
69. Phng L-K, Stanchi F, Gerhardt H (2013) **Filopodia are dispensable for endothelial tip cell guidance** *Development* **140**:4031–4040 <https://doi.org/10.1242/dev.097352>
70. Prasad GV, Coury LA, Finn F, Zeidel ML (1998) **Reconstituted aquaporin 1 water channels transport CO<sub>2</sub> across membranes** *J Biol Chem* **273**:33123–6 <https://doi.org/10.1074/jbc.273.50.33123>
71. Preston GM, Carroll TP, Guggino WB, Agre P (1992) **Appearance of water channels in *Xenopus* oocytes expressing red cell CHIP28 protein** *Science* **256**:385–7 <https://doi.org/10.1126/science.256.5055.385>
72. Rehn K, Wong KS, Balciunas D, Sumanas S (2011) **Zebrafish enhancer trap line recapitulates embryonic aquaporin 1a expression pattern in vascular endothelial cells** *Int J Dev Biol* **55**:613–618 <https://doi.org/10.1387/ijdb.103249kp>

73. Ridley AJ (2011) **Life at the Leading Edge** *Cell* **145**:1012–1022 <https://doi.org/10.1016/j.cell.2011.06.010>
74. Saadoun S, Papadopoulos MC, Hara-Chikuma M, Verkman AS. (2005) **Impairment of angiogenesis and cell migration by targeted aquaporin-1 gene disruption** *Nature* **434**:786–792 <https://doi.org/10.1038/nature03460>
75. Sato Y, Shigematsu M, Shibata-Kanno M, Maejima S, Tamura C, Sakamoto H (2023) **Aquaporin regulates cell rounding through vacuole formation during endothelial-to-hematopoietic transition** *Development* **150** <https://doi.org/10.1242/dev.201275>
76. Sauter L, Affolter M, Belting H-G (2017) **Distinct and redundant functions of Esam and VE-cadherin during vascular morphogenesis** *Development* **144**:1554–1565 <https://doi.org/10.1242/dev.140038>
77. Siekmann AF, Affolter M, Belting H-G (2013) **The tip cell concept 10 years after: New players tune in for a common theme** *Exp Cell Res* **319**:1255–1263
78. Son S, Kang JH, Oh S, Kirschner MW, Mitchison TJ, Manalis S (2015) **Resonant microchannel volume and mass measurements show that suspended cells swell during mitosis** *J Cell Biol* **211**:757–763 <https://doi.org/10.1083/jcb.201505058>
79. Soria LR, Fanelli E, Altamura N, Svelto M, Marinelli RA, Calamita G (2010) **Aquaporin-8-facilitated mitochondrial ammonia transport** *Biochem Biophys Res Commun* **393**:217–221 <https://doi.org/10.1016/j.bbrc.2010.01.104>
80. Stewart MP, Helenius J, Toyoda Y, Ramanathan SP, Muller DJ, Hyman AA (2011) **Hydrostatic pressure and the actomyosin cortex drive mitotic cell rounding** *Nature* **469**:226–230 <https://doi.org/10.1038/nature09642>
81. Stroka KM, Jiang H, Chen S-H, Tong Z, Wirtz D, Sun SX, Konstantopoulos K (2014) **Water permeation drives tumor cell migration in confined microenvironments** *Cell* **157**:611–623 <https://doi.org/10.1016/j.cell.2014.02.052>
82. Talbot K, Kwong RWM, Gilmour KM, Perry SF (2015) **The water channel aquaporin-1a1 facilitates movement of CO<sub>2</sub> and ammonia in zebrafish (*Danio rerio*) larvae** *J Exp Biol* **218**:3931–3940 <https://doi.org/10.1242/jeb.129759>
83. Thisse C, Thisse B (2007) **High-resolution in situ hybridization to whole-mount zebrafish embryos** *Nat Protoc* **3**:59–69 <https://doi.org/10.1038/nprot.2007.514>
84. Tingaud-Sequeira A, Calusinska M, Finn RN, Chauvigné F, Lozano J, Cerdà J (2010) **The zebrafish genome encodes the largest vertebrate repertoire of functional aquaporins with dual paralogy and substrate specificities similar to mammals** *BMC Evol Biol* **10** <https://doi.org/10.1186/1471-2148-10-38>
85. Wakayama Y, Fukuhara S, Ando K, Matsuda M, Mochizuki N (2015) **Cdc42 Mediates Bmp-Induced Sprouting Angiogenesis through Fmnl3-Driven Assembly of Endothelial Filopodia in Zebrafish** *Dev Cell* **32**:109–122 <https://doi.org/10.1016/j.devcel.2014.11.024>

86. Wang Y, Cohen J, Boron WF, Schulten K, Tajkhorshid E (2007) **Exploring gas permeability of cellular membranes and membrane channels with molecular dynamics** *J Struct Biol* **157**:534–544 <https://doi.org/10.1016/j.jsb.2006.11.008>
87. Wu DQ, Yang ZF, Wang KJ, Feng XY, Lv ZJ, Li Y, Jian ZX (2017) **AQP8 inhibits colorectal cancer growth and metastasis by down-regulating PI3K/AKT signaling and PCDH7 expression** *Am J Cancer Res* **8**:266–279
88. Zeidel ML, Ambudkar SV, Smith BL, Agre P (1992) **Reconstitution of functional water channels in liposomes containing purified red cell CHIP28 protein** *Biochemistry* **31**:7436–7440 <https://doi.org/10.1021/bi00148a002>

## Editors

Reviewing Editor

**Shane Herbert**

University of Manchester, Manchester, United Kingdom

Senior Editor

**Didier Stainier**

Max Planck Institute for Heart and Lung Research, Bad Nauheim, Germany

## Reviewer #1 (Public Review):

Summary:

This paper details a study of endothelial cell vessel formation during zebrafish development. The results focus on the role of aquaporins, which mediate the flow of water across the cell membrane, leading to cell movement. The authors show that actin and water flow together drive endothelial cell migration and vessel formation. If any of these two elements are perturbed, there are observed defects in vessels. Overall, the paper significantly improves our understanding of cell migration during morphogenesis in organisms.

Strengths:

The data are extensive and are of high quality. There is a good amount of quantification with convincing statistical significance. The overall conclusion is justified given the evidence.

Weaknesses:

There are two weaknesses, which if addressed, would improve the paper.

(1) The paper focuses on aquaporins, which while mediates water flow, cannot drive directional water flow. If the osmotic engine model is correct, then ion channels such as NHE1 are the driving force for water flow. Indeed this water is shown in previous studies. Moreover, NHE1 can drive water intake because the export of H<sup>+</sup> leads to increased HCO<sub>3</sub> due to the reaction between CO<sub>2</sub>+H<sub>2</sub>O, which increases the cytoplasmic osmolarity (see Li, Zhou and Sun, *Frontiers in Cell Dev. Bio.* 2021). If NHE cannot be easily perturbed in zebrafish, it might be of interest to perturb Cl channels such as SWELL1, which was recently shown to work together with NHE (see Zhang, et al, *Nat. Comm.* 2022).

(2) In some places the discussion seems a little confusing where the text goes from hydrostatic pressure to osmotic gradient. It might improve the paper if some background is given. For example, mention water flow follows osmotic gradients, which will build up hydrostatic

pressure. The osmotic gradients across the membrane are generated by active ion exchangers. This point is often confused in literature and somewhere in the intro, this could be made clearer.

<https://doi.org/10.7554/eLife.98612.1.sa2>

#### **Reviewer #2 (Public Review):**

##### Summary:

Directional migration is an integral aspect of sprouting angiogenesis and requires a cell to change its shape and sense a chemotactic or growth factor stimulus. Kondrychyn I. et al. provide data that indicate a requirement for zebrafish aquaporins 1 and 8, in cellular water inflow and sprouting angiogenesis. Zebrafish mutants lacking *aqp1a.1* and *aqp8a.1* have significantly lower tip cell volume and migration velocity, which delays vascular development. Inhibition of actin formation and filopodia dynamics further aggravates this phenotype. The link between water inflow, hydrostatic pressure, and actin dynamics driving endothelial cell sprouting and migration during angiogenesis is highly novel.

##### Strengths:

The zebrafish genetics, microscopy imaging, and measurements performed are of very high quality. The study data and interpretations are very well-presented in this manuscript.

##### Weaknesses:

Some of the findings and interpretations could be strengthened by additional measurements and further discussion. Also, a better comparison and integration of the authors' findings, with other previously published findings in mice and zebrafish would strengthen the paper.

<https://doi.org/10.7554/eLife.98612.1.sa1>

#### **Reviewer #3 (Public Review):**

##### Summary:

Kondrychyn and colleagues describe the contribution of two Aquaporins *Aqp1a.1* and *Aqp8a.1* towards angiogenic sprouting in the zebrafish embryo. By whole-mount in situ hybridization, RNAscope, and scRNA-seq, they show that both genes are expressed in endothelial cells in partly overlapping spatiotemporal patterns. Pharmacological inhibition experiments indicate a requirement for VEGFR2 signaling (but not Notch) in transcriptional activation.

To assess the role of both genes during vascular development the authors generate genetic mutations. While homozygous single mutants appear less affected, *aqp1a.1;aqp8a.1* double mutants exhibit severe defects in EC sprouting and ISV formation.

At the cellular level, the aquaporin mutants display a reduction of filopodia in number and length. Furthermore, a reduction in cell volume is observed indicating a defect in water uptake.

The authors conclude, that polarized water uptake mediated by aquaporins is required for the initiation of endothelial sprouting and (tip) cell migration during ISV formation. They further propose that water influx increases hydrostatic pressure within the cells which may facilitate actin polymerization and formation membrane protrusions.

#### Strengths:

The authors provide a detailed analysis of Aqp1a.1 and Aqp8a.1 during blood vessel formation *in vivo*, using zebrafish intersomitic vessels as a model. State-of-the-art imaging demonstrates an essential role in aquaporins in different aspects of endothelial cell activation and migration during angiogenesis.

#### Weaknesses:

With respect to the connection between Aqp1/8 and actin polymerization/filopodia formation, the evidence appears preliminary and the authors' interpretation is guided by evidence from other experimental systems.

<https://doi.org/10.7554/eLife.98612.1.sa0>